

Therapeutic Plasma Exchange and Other Extracorporeal Treatments in Management of Kidney and Related Diseases

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CHAPTER OUTLINE

CLINICAL RELEVANCE AND HISTORICAL PERSPECTIVE, 1900

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CONCLUSION, 1917

KEY POINTS

- Therapeutic plasma exchange (TPE) and other extracorporeal separation techniques can be effective in removing large toxins, pathogenic antibodies, and neutralizing targeted substances and is often combined with immunosuppression.
- TPE is considered first-line treatment in the management of patients with pulmonary hemorrhage in the setting of anti-GBM disease or vasculitis.
- TPE is indicated in patients with anti-GBM disease (without advanced kidney failure) and considered in rapidly progressive glomerulonephritis (with advanced renal failure).
- The evidence for using TPE in cast nephropathy is controversial, but TPE is indicated in patients with hyperviscosity syndrome.
- TPE is indicated for catastrophic anti-phospholipid antibody syndrome.
- TPE is first-line therapy in thrombocytopenic purpura (low ADAMTS-13) levels and factor H autoantibody-mediated thrombocytopenic purpura.
- In renal transplantation, TPE is indicated for patients with antibody-mediated rejection, desensitization, and recurrent FSGS post transplantation.
- TPE is indicated in acute liver failure, often combined with other therapies.
- TPE is effective in nonkidney diseases such as fulminant Wilson disease, familial hypercholesterolemia, and neurologic diseases including Guillain-Barré syndrome, demyelinating polyneuropathies, and myasthenia gravis.
- The only drug for which TPE is indicated is ticlopidine.
- Lipoprotein apheresis is indicated for familial hypercholesterolemia.

CLINICAL RELEVANCE AND HISTORICAL PERSPECTIVE

The word “apheresis” is derived from the Greek word that means “taking away” or “separation.” Therapeutic apheresis encompasses all modalities where a substance is removed from the body. A summary of the conditions and techniques is periodically reviewed and updated.¹ Therapeutic plasma exchange (TPE) is an extracorporeal therapeutic procedure in which blood from a patient is separated into plasma and other blood components. The plasma is removed and replaced with another solution. The term “plasmapheresis” refers to the same procedure but does not use a replacement solution. In this chapter, TPE is used instead of plasmapheresis according to guidelines of the American Society for Apheresis.¹ It is unclear when the notion of therapeutic removal of blood components first originated, but it was flourishing even before Hippocrates in the 5th century BC. Bloodletting to remove evil humors was a commonplace medical practice, in part due to a lack of understanding of disease processes and the paucity of effective therapies. By the Middle Ages, surgeons and barbers were specializing in this bloody and often painful practice and, even as late as the 19th century, bloodletting was used for nearly every infectious and malignant malady afflicting patients in the United States and Europe.² Even today, bloodletting with the use of leeches (hirudotherapy) is a U.S. Food and Drug Administration–approved technique for nonoperative tissue salvage in plastic surgery patients. The first true TPE involving the removal of the “bad blood” and replacement with a clean solution was performed in 1914 by Abel, Rowntree, and Turner at the Johns Hopkins Hospital. The procedure was called “vivi-diffusion” and demonstrated the principle that the blood of a living animal could be dialyzed outside the body and then returned to the circulation.³ In 1960, Schwab and Fahey performed the first therapeutic manual plasmapheresis to reduce elevated globulin levels in a patient with Waldenstrom macroglobulinemia.⁴

TPE permits removal of high-molecular-weight substances such as antibodies, immune complexes, and toxic proteins. When used in combination with other therapies, TPE has been shown to improve patient outcomes in specific diseases. Although randomized controlled trials (RCTs) are often lacking, TPE has a role in the treatment of several kidney diseases, including antglomerular basement membrane (GBM) antibody disease, rapidly progressive glomerulonephritis (RPGN), thrombotic microangiopathies (TMA), and kidney transplantation. In addition, TPE is first-line therapy for nonkidney diseases such as Guillain-Barré syndrome, demyelinating polyradiculopathy, and familial hypercholesterolemia (FH). The procedure is generally well tolerated, with few side effects, but usually requires central venous catheter placement. In the early days, the utility of TPE was based on anecdotal or uncontrolled studies but, in recent years, the number of clinical indications for this procedure has been growing. However, only a few clinical conditions have been rigorously studied with prospective and RCTs, and decisions for the implementation of TPE still rest on anecdotal and uncontrolled studies in most cases. This

chapter reviews the technical aspects of the TPE process and the major conditions for which it has therapeutic benefit.

GENERAL PRINCIPLES OF THERAPEUTIC PLASMA EXCHANGE AND TECHNICAL ASPECTS/COMPLICATIONS

The mechanism(s) for clinical improvement of kidney diseases with TPE depend on the pathophysiology of the underlying disease. TPE should be considered when the pathogenic factor is a high-molecular-weight substance or when the patient is deficient in a plasma component. However, hemodialysis and hemofiltration are more efficient procedures for the removal of small molecules and toxins with large volumes of distribution. The success of TPE in treating any disease characterized by accumulation of toxic proteins or antibodies depends on two variables: 1. the rate of production of the abnormal protein or substance; and 2. the efficiency of removal with TPE (Fig. 65.1). This balance will determine whether an abnormal component can be removed sufficiently rapidly to provide clinical benefit. This occurs in parallel to the end-organ damage, so the ultimate benefit of the procedure depends on the efficiency of toxin reduction and rate of end-organ damage. As a result, TPE is rarely used in isolation but is most often used in combination with immunosuppressive strategies to decrease production and reduce inflammation. In addition to the removal of toxic proteins or replacement of deficient ones (Table 65.1) with plasma exchange, other benefits may include reversal of impaired splenic function to remove immune complexes, removal of fibrinogen, replacement of humoral factors, changes in natural killer cell numbers and activity, decline in B cells, an increase in T cells, an increase in T-suppressor cell function, and a shift from a T helper 2 (Th2)-predominant pattern to a Th1-predominant pattern.⁵

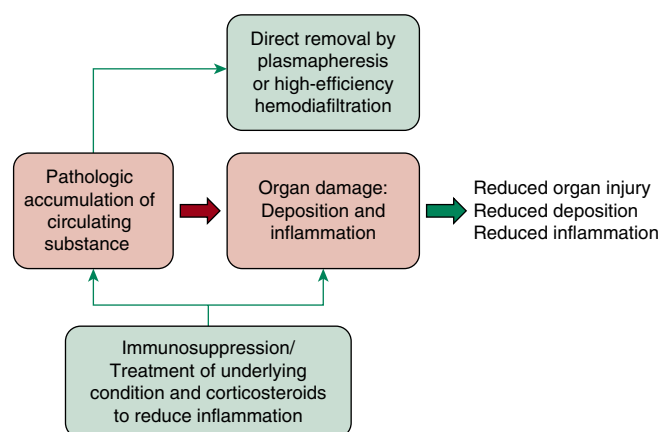


Fig. 65.1 Rationale and role of therapeutic plasma exchange for reducing end-organ injury. Pathologic conditions result in accumulation of toxic proteins, antibodies, or other molecules that induce tissue damage directly and through the associated inflammatory response. A two-pronged approach to slow or arrest this process is to 1. remove toxic molecules when feasible (top green box) and 2. suppress production of these substances with appropriate immunosuppressive medication (lower green box). Corticosteroids are both immunosuppressive and anti-inflammatory.

Table 65.1 Pathologic Factors Removed by Therapeutic Plasma Exchange

Autoantibodies
Immune complexes
Myeloma proteins
Cryoglobulin
Complement products
ADAMTS-13 (metalloproteinase)
Lipoproteins
Protein-bound toxins

TECHNICAL ASPECTS AND COMPLICATIONS

The TPE technique involves the withdrawal of venous blood through a needle or central catheter to separate the plasma from blood cells by centrifugation or membrane filtration and reinfuse the cells, plus autologous plasma or another replacement solution. For most conditions, the aim of the procedure is the removal of pathologic autoantibodies or toxins; the initial treatment goal is to exchange 1 to 1.5 times the plasma volume per TPE procedure. This will lower plasma macromolecule levels by 60% to 75%, respectively. A formula that can be used to estimate plasma volume in an adult is as follows⁶:

$$\text{Estimated plasma volume (in L)} = 0.07 \text{ weight (in kg)} \times (1 - \text{hematocrit [Hct]})$$

The ultimate clinical success of the procedure depends on the balance of production and removal of the offending molecule. The time required to suppress abnormal protein production can take several weeks, which is why TPE protocols often require daily apheresis (or near-daily) for prolonged times.

PLASMA SEPARATION TECHNIQUES

The most common method for separating the plasma from the blood during a TPE procedure is centrifugation (Fig. 65.2). Alternatively, TPE can be accomplished using a membrane filter and standard hemodialysis equipment. However, the centrifugal methods are more efficient.⁷ Potential disadvantages of membrane filtration include activation of complement and leukocytes by the artificial membrane and the need for a central catheter to obtain adequate blood flow rates. If severe kidney failure is present, and dialysis is required, the membrane filtration method can be done in combination with conventional hemodialysis. Both centrifugation and membrane filtration are safe and efficient TPE techniques; the main differences are the cost and expertise needed to operate.

The centrifugation method uses centrifugal force to separate whole blood into plasma and cellular fractions according to their density. The centrifugation process can be intermittent or continuous. The blood is pumped from the patient at a flow rate of up to 100 mL/min into the processing unit, which consists of a bell-shaped bowl that rotates at high speed. The denser cellular blood components

are centrifuged against the lateral walls, and the plasma is removed through a central outlet on the top of the bowl. In intermittent centrifugation, a single needle can be used and each cycle removes about 500 to 700 mL of plasma; therefore the process needs to be repeated five or six times to achieve the goal of 2.5 to 4.0 L (1–1.5 plasma volumes) and usually takes >4 hours. In the continuous flow centrifugation system, the blood is pumped continuously into a rapidly rotating bowl, where plasma and cells are separated. Plasma is removed at a specified rate, and the cells plus replacement fluid are returned to the patient in a continuous manner. This method is faster and is more suitable for hemodynamically unstable patients; however, it is more costly and requires two venipunctures or insertion of a dual-lumen central venous catheter.

VENOUS ACCESS

TPE requires a reliable venous access. The clinical scenario, especially the possibility for long-term venous access, and the type of TPE being used are important factors to consider when deciding on peripheral or central venous access. A peripheral vein allows a maximum flow of up to about 50 to 90 mL/min, so a single venous access is adequate for intermittent centrifugation, whereas continuous centrifugation techniques will require two venous access sites. For short-term procedures, this may be adequate but loss of venous access from recurrent intravenous catheters and phlebotomy in chronically ill patients is a major problem. If a long-term TPE is planned (>1–2 weeks), then a central venous catheter is required, preferably one that is tunneled beneath the skin; this generally reduces the likelihood of infection and allows for better catheter performance. When the membrane filtration technique is used, a central venous catheter is necessary to sustain blood flows >70 mL/min. Central venous access can be achieved through the femoral, internal jugular, or subclavian vein; the femoral vein should be avoided if the treatment of the patient will be ambulatory.⁸ In patients who require lifelong therapy, such as LDL apheresis, either an arteriovenous (AV) fistula or AV graft should be considered.

Central venous catheters have numerous long-term complications, including catheter thrombosis, catheter infection, pneumothorax, central vein thrombosis, and vein stenosis. See Chapter 62 regarding the catheters used for maintenance hemodialysis.

ANTICOAGULATION

To prevent clotting of the extracorporeal circuit, TPE requires anticoagulation. For centrifugation procedures, the acid-citrate-dextrose (ACD) solution (one-ninth the volume of solute/volume of solution), given as a continuous intravenous infusion, is the most frequently used anticoagulant. The infusion rate is adjusted according to the blood flow rate (target ratio ranges 1:10–1:25). When the venous flux and infusion rate of citrate are slow, there is an increased risk for catheter clotting. In this case, heparin (if not contraindicated) can be used alone or in combination with citrate.

For membrane filtration TPE procedures, the use of standard unfractionated heparin is preferred; the required

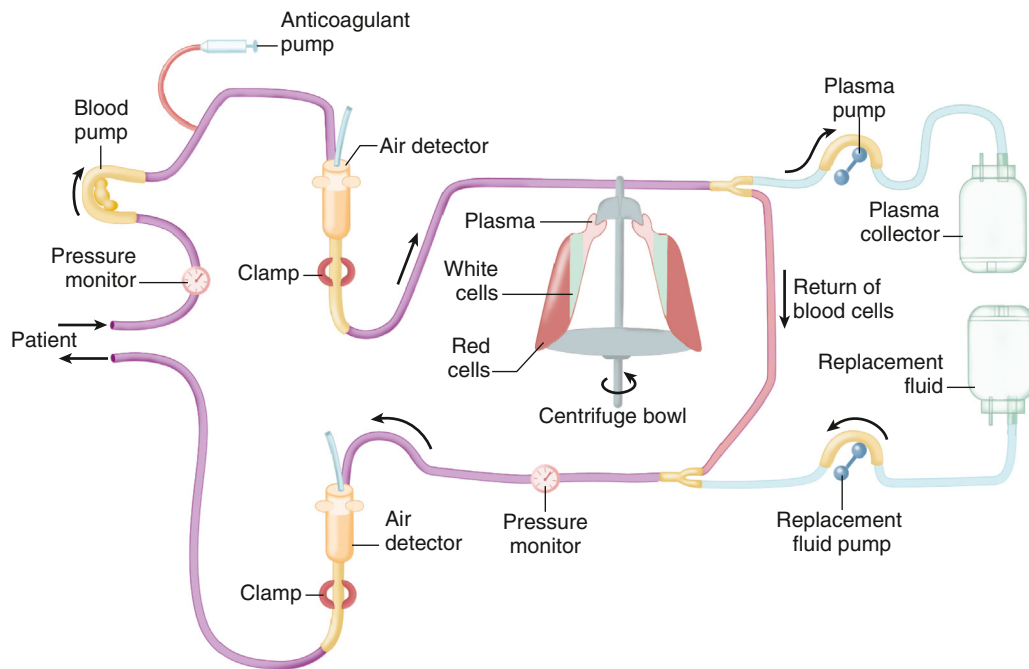


Fig. 65.2 Centrifugal separator for plasma exchange. Blood is pumped into the separator container. As the centrifuge revolves, different blood components are separated into discrete layers, which can be harvested separately. Plasma is pumped out of the centrifuge into a collection chamber. Red blood cells, leukocytes, and platelets are returned to the donor, along with replacement fluid. (From Madore F, Lazarus JM, Brady HR. Therapeutic plasma exchange in kidney diseases. *J Am Soc Nephrol.* 1996;7:367–386.)

dose of heparin is about twice that needed for hemodialysis because a significant amount of infused heparin is removed along with the plasma. However, heparin may enhance systemic anticoagulation more than expected because of the additional effect of dilution of clotting factors by the nonplasma replacement solutions. The initial loading dose of heparin (40 U/kg) is usually administered intravenously, followed by a continuous infusion (20 U/kg per hour) adjusted to maintain an adequate anticoagulation in the circuit.^{9,10}

For patients who are receiving standard oral anticoagulation—warfarin or one of the direct-acting oral anticoagulants—additional low-dose anticoagulation with regional citrate or heparin should be added to facilitate the treatment and prevent clotting during the procedure. The heparin dose can usually be reduced by at least 50% in this situation.¹¹ In critically ill patients with coagulation abnormalities, the use of regional citrate is recommended.¹² Hirudin and lepirudin (thrombin inhibitors) are effective and relatively safe alternatives for patients with increased risk for thrombosis but have contraindications for heparin administration (e.g., heparin-induced thrombocytopenia and thrombosis).^{13,14}

REPLACEMENT FLUID

The choice of replacement fluids includes 5% albumin, fresh-frozen plasma (FFP) (or other plasma derivatives such as cryosupernatant), crystalloids (e.g., 0.9% saline and Ringer lactate), and synthetic plasma expanders (e.g., hydroxyethyl starch [HES]).¹⁵ Albumin is the most commonly used solution in TPE and is generally combined with 0.9% saline on a 50:50 basis. Albumin does not contain

calcium or potassium and also lacks coagulation factors and immunoglobulins. It is safe and has never been associated with transmission of hepatitis or HIV.¹⁶ FFP contains complement and coagulation factors and is the replacement fluid of choice in patients with thrombocytopenic purpura (TTP) because the infusion of normal plasma may contribute to the replacement of the deficient plasma factor ADAMTS13. Plasma may also be preferable in patients at risk of bleeding (e.g., those with liver disease, disseminated intravascular coagulation, after a renal biopsy) or those requiring intensive therapy (e.g., daily exchanges for several weeks) because frequent replacements with albumin solution will eventually result in post-TPE coagulopathy and a net loss of immunoglobulins. The disadvantages of using FFP include the risk of viral disease transmission and citrate overload.

Colloidal starch has been used as a replacement solution. It is well tolerated, drug interaction is minimal, the potential for disease transmission is absent, and it is cost-effective when compared with human albumin.¹⁷ HES is a polysaccharide colloid used as an agent for plasma volume expansion and for enhancement of granulocyte yields during leukapheresis. Its pharmacology and safety have been well studied and, when used in moderate amounts in humans and animals, HES produces relatively minor changes in coagulation tests, and overt bleeding rarely occurs. In TPE procedures, the side effects of HES on coagulation factors and tests are comparable with those of other replacement fluids. HES can be safely used in the first two sessions of therapy in patients with an albumin level above 30 g/dL to perform short-term TPE;¹⁸ longer HES exposure (130 L within 20 months) can lead to symptomatic effects (e.g., sensory polyneuropathy

and weight loss) and diffuse tissue infiltration with HES-laden foamy macrophages.¹⁹ Excessive HES exposure in patients with impaired renal function can result in acquired lysosomal storage disease, so HES should be avoided in chronic TPE procedures. Contraindications for using starches in TPE are congestive heart failure, renal or liver failure, coagulopathy, hyperviscosity, allergic reactions to starches, pregnancy, breastfeeding, and pediatric patients.²⁰

COMPLICATIONS

TPE is a well-tolerated and relatively safe procedure and, although adverse events are common, death is rare, occurring in <0.1% of all procedures.²¹ There are fewer adverse reactions when albumin is administered as volume replacement than when using FFP (1.4% vs. 20%). The high-risk criteria for adverse reactions associated with TPE are unstable vital signs, hypotension, active bleeding, severe bronchoconstriction, severe anemia, pregnancy, and conditions requiring continuous nursing support.²² Table e-65.1 summarizes the most common complications related to the procedure.

The Swedish registry of therapeutic apheresis has reported more than 14,000 procedures from 1996 to 1999; adverse events occurred in 4.2% of procedures, and no fatalities were reported. Of all the apheresis procedures, 1% had to be interrupted due to an adverse event. The most common adverse effects reported were paresthesias (0.52%), hypotension (0.5%), urticaria (0.34%), shivering, and nausea. Kiprof and associates,²³ in another report of 17,940 procedures performed on 3583 patients, had an occurrence rate of adverse events in 3.9% of all procedures. The following adverse reactions were documented: reactions related to citrate toxicity (3%), vasovagal reactions and hypotension (0.5%), vascular access-related complications (0.15%), reactions related to FFP (0.12%), hepatitis B from FFP (0.06%), arrhythmias (0.01%), hemolysis due to inappropriate dilution of 25% albumin (0.01%), and one death (from underlying disease) during a TPE procedure (0.006%). No significant bleeding complications were observed, and patients receiving FFP had significantly higher rates of adverse reactions than patients receiving other exchange fluids. More recently, a retrospective cohort of 435 children treated with 1201 TPE procedures found 152 adverse events with pruritis/urticaria in 7% and hypotension in 1.17%. One child developed disseminated cryptococcosis neoformans, three children developed toxic epidermal necrolysis, one had anaphylactic shock, but there were no deaths.²⁴

One of the most frequent complications of TPE is the hypocalcemia related to citrate infusion as an anticoagulant for the extracorporeal system or in the FFP administered as a replacement fluid. Citrate binds to free calcium to form soluble calcium citrate, thereby lowering the free but not the total serum calcium concentration. Hypocalcemia may manifest with perioral and distal extremity paresthesias. Symptoms can be prevented and reduced by administration of intravenous or oral calcium if the TPE therapy will last longer than 1 hour. The administration of oral calcium carbonate or addition of calcium gluconate to the return fluid is a useful maneuver to prevent hypocalcemia.²⁵ The incidence of hypocalcemic symptoms is lowered with the

prophylactic administration of calcium; without calcium prophylaxis, the incidence of symptoms was 9.1% (6 in 66 treatments), whereas with calcium prophylaxis the incidence was reduced to 1% (6 in 633 treatments). Marques and colleagues²⁶ have reported an incidence of hypocalcemia of 3% when calcium gluconate is infused in 5% albumin.

Another complication of citrate administration is the development of metabolic alkalosis, but serum concentrations of bicarbonate higher than 35 mEq/L are rarely seen. Risk factors more frequently associated with this complication are use of FFP and low GFR (e.g., patients with TTP); this is because the excess of citrate generates bicarbonate, the excretion of which is limited by impaired kidney function. Replacement regimens using saline and albumin solutions can result in a 25% reduction in the plasma potassium concentration in the postapheresis period, which can be minimized by adding 4 mEq potassium/L to the replacement solution. Hypokalemia is also a consequence of metabolic alkalosis.²⁶

TPE can lead to a reduction in blood pressure that is usually due to a decrease in intravascular volume. Because the volume of extracorporeal whole blood is higher with intermittent centrifugation techniques, hypotension episodes are more common than with continuous modalities. Hypotension can also occur in response to complement-mediated reactions to the membrane filter or sensitivity to the ethylene oxide that is used to sterilize the membrane. Intravenous administration of FFP can rarely result in anaphylactoid reactions, themselves rarely resulting in death. FFP reactions are most often characterized by fever, rigors, urticaria, wheezing, and hypotension.

The development of dyspnea suggests the presence of pulmonary edema due to fluid overload; noncardiogenic edema can rarely occur as a component of anaphylactoid reactions. Rarely, massive pulmonary emboli can develop if the blood components that are reinfused are not adequately anticoagulated.

Plasma exchange with albumin replacement produces a predictable decrease in clotting factors that may predispose to bleeding (Table e-65.2). A single plasma volume exchange increases the prothrombin time by 30% and the partial thromboplastin time by 100%; these changes return toward normal within several hours, but with repeated TPE sessions, these abnormalities can persist. In reported studies, the most significant change is in the fibrinogen. Keller and colleagues²⁷ have reported that fibrinogen levels decrease to 25% of pre-TPE levels and recover to baseline after 2 to 3 days. Therefore 1 L or more of FFP (3–4 U/L) should be substituted as the replacement fluid each week or sooner in patients at risk for bleeding. Thrombocytopenia is also a consequence of plasma removal, with larger volumes removed associated with greater platelet loss; the mean platelet reduction following a TPE procedure ranges from 9.4% to 52.6%. Clinical bleeding associated with TPE is rarely reported, and when TPE-related hemorrhage is present, it is more likely a consequence of thrombocytopenia or inadequate heparin neutralization.^{28,29}

Removal of immunoglobulins and complement can result in immunodeficiency, which can exacerbate immunosuppressive effects of other therapeutic agents (e.g., corticosteroids, cyclophosphamide, and rituximab). However, in an RCT of TPE in patients with

lupus nephritis, TTP, or multiple myeloma, patients receiving TPE were not more prone to infection than the other patients.³⁰ Nevertheless, repeated apheresis treatments with albumin replacement will deplete the patient's reserve of immunoglobulins for several weeks. If an infection occurs, a single infusion of intravenous immunoglobulin (IVIg) (100–400 mg/kg) will restore the plasma immunoglobulin concentration toward normal. Although estimates for the risk of viral transmission by the use of FFP are low, the large volumes from multiple donors increase the risk in patients receiving long-term TPE therapy. Use of large-volume plasma units collected from a single donor and the use of hepatitis B vaccine may reduce the risk of the patient acquiring a virally transmitted infection.

Substantial drug removal by TPE occurs for those drugs that are highly protein bound and therefore primarily limited to the vascular space. Among drugs used to treat kidney diseases, prednisone is not substantially removed, whereas cyclophosphamide and azathioprine are removed to some extent. This potential problem can be circumvented by administering the drug after a plasma exchange treatment.

Flushing, hypotension, abdominal cramping, and other gastrointestinal symptoms have been reported during TPE in patients receiving angiotensin-converting enzyme (ACE) inhibitors. In one report of 299 consecutive patients undergoing TPE, these atypical symptoms occurred in all 14 patients receiving an ACE inhibitor versus only 7% of those not treated with this medication.³¹ The administration of an ACE inhibitor may prolong the half-life of bradykinin, thus permitting the attainment of a clinically significant concentration in the plasma, so it is recommended to withhold this type of drug for 24 hours before TPE.³²

In summary, TPE is a relatively safe procedure, with most of complications being mild and reversible. However, moderate to severe reactions, even death, can occur, especially in patients with severe primary diseases.

OTHER EXTRACORPOREAL TREATMENT MODALITIES

The double-filtration plasmapheresis (DFPP; also called “cascade filtration”) is another variation of membrane plasmapheresis. In this technique, the plasma that has been separated by the membrane flows again through membranes with different pore diameters and filtration and adsorption characteristics, high-molecular-weight proteins are discarded, and low-molecular-weight substances, including valuable albumin, are returned to the patient. A small amount of substitution fluid, such as albumin, may be added.³³

In recent years, these basic techniques have been modified (such as high volume) and/or coupled to other separation modalities. Cytapheresis is the removal of leukocytes or platelets in hematologic conditions with hyperleukocytosis or thrombocytosis. Erythrocytapheresis (red blood cell exchange) can also be performed for sickle cell crisis; in this setting, the goal is the removal of more than 50% of hemoglobin S and replacement with normal allogenic red cells.

When the plasma filtration is carried out above a normal physiologic temperature, the process is called “thermo-filtration.” This procedure is performed on patients with severe dyslipidemia, whereas cryofiltration is when the procedure is done with the temperature below normal and is used to remove immunoglobulin and immune complexes. Alternatively, plasma adsorption and immunoabsorption procedures use affinity columns for processing the separated plasma. Adsorption columns, such as protein A columns, remove IgG antibodies and immune complexes; chemical affinity columns, such as dextran sulfate, have a negative charge and are used to remove antibodies or other positively charged plasma substances, such as LDLs and very-low-density lipoproteins (VLDLs).³⁴ The Molecular Adsorbent Recirculating System (MARS) is a highly specialized combination of plasma separation, protein adsorption, and continuous dialysis used in some centers for treatment of acute liver failure (see Fig. 65.3).

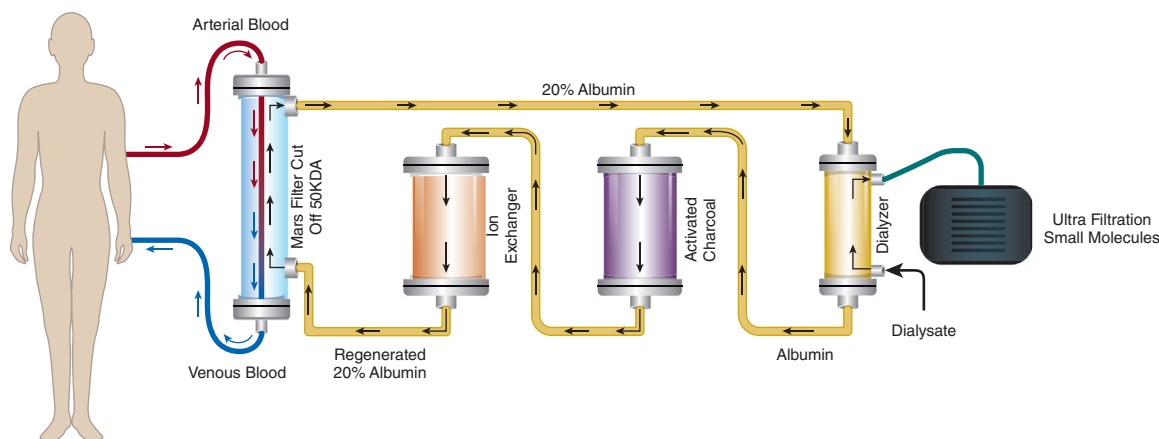


Fig. 65.3 Molecular Adsorbent Recirculating System (MARS). MARS provides enhanced toxin removal by combining traditional removal of water-soluble toxins through conventional hemodialysis and albumin-bound toxins by linking an albumin circuit to remove protein-bound toxins. Blood is pumped from the patient into the MARS flux dialyzer that has a 10% to 20% albumin dialysate flowing countercurrent at ~600 mL/min. The MARS dialyzer is porous for molecules up to 50kDa, so it restricts growth factors and albumin from freely crossing the membrane, but smaller protein-bound toxins filter through and the albumin dialysate has capacity to bind the filtered toxins. The albumin dialysate is then sequentially circulated through a charcoal filter and ion exchange resin before the “cleaned” albumin dialysate is returned to the patient. The standard continuous renal replacement therapy filter is used for ultrafiltration and removal of water-soluble toxins.

PLASMAPHERESIS USE IN KIDNEY DISEASE

The clinical indications of TPE for kidney diseases according to category are summarized in Table 65.2.

ANTI-GLOMERULAR BASEMENT MEMBRANE DISEASE

Anti-glomerular basement membrane (anti-GBM) disease is a disorder in which circulating antibodies are directed against the non-collagen (NC)-1 domain of the α_3 chain of type IV collagen, resulting in RPGN. Goodpasture syndrome

is classically defined by the triad of pulmonary hemorrhage, RPGN, and circulating anti-GBM antibodies. Table 65.3 summarizes renal outcome studies discussed later in patients with anti-GBM-mediated kidney failure.

The epidemiology of anti-GBM disease reveals that there may be seasonal variation and temporal clustering that suggest environmental factors may trigger disease in genetically susceptible individuals.³⁵ Anti-GBM disease was associated with respiratory tract infections and prodromal fever in 140 patients in China.³⁶ Smoking has also been linked to anti-GBM disease.³⁷ Cigarette smoking and lung hemorrhage in glomerulonephritis caused by autoantibodies to the glomerular basement membrane increases the risk of patients presenting with pulmonary hemorrhage. These observations suggest that inflammation or damage to the alveoli alters the conformation of Col IV leading to subsequent antibody formation and subsequent antibody-mediated injury and alveolar hemorrhage. More than 90% of patients have circulating anti-GBM antibodies, and the titer of circulating antibodies correlates with disease activity. Approximately 60% to 70% of patients will have pulmonary disease in addition to RPGN and, rarely, a patient may present with pulmonary hemorrhage and no renal involvement.³⁸

Before the use of TPE and immunosuppression, the mortality rate exceeded 90%, with a mean survival time of <4 months. The use of TPE was introduced for the treatment of anti-GBM disease in 1975, in an adjunct manner with immunosuppressives.³⁹ Immunosuppression and plasma-exchange in the treatment of Goodpasture syndrome has remained the prototype for treatment. Numerous uncontrolled studies and case series published thereafter have suggested the beneficial effect of TPE on overall survival and renal survival. The low incidence, sudden

Table 65.2 Summary of Kidney Diseases Treated With Plasmapheresis

Disease	Category
Antiglomerular basement membrane disease	I
Rapidly progressive glomerulonephritis	II
Hemolytic uremic syndrome	III
TTP	I
Renal transplant rejection	IV
Desensitization for renal transplantation	II
Recurrent FSGS	III
Cryoglobulinemia	II
Systemic lupus erythematosus	III

Category I, Standard primary therapy; Category II, supportive therapy; Category III, when the evidence of benefit is unclear; Category IV, when there is no current evidence of benefit or for research protocols; FSGS, focal and segmental glomerulosclerosis; TTP, thrombotic thrombocytopenic purpura.

Table 65.3 Renal Recovery According to Initial Creatinine Concentration in Treated Patients With Anti-Glomerular Basement Membrane Antibody Disease

Study, Year (Reference)	Patients (No.)	Patients With Independent Renal Function at 1 Year		Treatment
		Initial Cr Concentration <600 mol/L (<6.8 mg/dL)	Initial Cr Concentration $\geq$ 600 mol/L ($\geq$ 6.8 mg/dL) or dialysis dependent	
Bouget et al., 1990	13	50%	0%	Most patients received PE
Herody et al., 1993	29	93%	0%	Most patients received PE
Merkel et al., 1994	32	64%	3%	25 patients received PE
Andrews et al., 1995	15	NA	7%	All patients had Cr concentration $\geq$ 600 mmol/L, only 8 patients received treatment
Daly et al., 1996	40	20%	0%	23 patients received PE
Levy et al., 2001	71	94%	15%	All patients received PE, C, and CFM
Saurina et al., 2003	32	71%	18%	24 patients received treatment with C, CFM and PE
Cui et al., 2011	176	25%	3%	76 patients C, CFM and PE, 59 PE + C, 41C
Huart et al., 2016	122	40%	2.4%	101 patients PE/C/CFM, 12 PE/C or PE/CGM
Zahir et al., 2021	48	4%	8%	31 patients PE/C/CFM, 5 PE/C, 5 C/CFM, 12 with no immunosuppression

C, Corticosteroids; CFM, cyclophosphamide; Cr, Creatinine; PE, plasmapheresis.

onset, and severity of disease have made it challenging to conduct RCTs. Currently, with the combination of TPE, corticosteroids, and cyclophosphamide, the mortality rate has been reduced to <20%. The role of TPE in anti-GBM disease involves the rapid removal of pathogenic antibodies, whereas cyclophosphamide is essential to reduce ongoing antibody production. The use of high-dose steroids targets the acute inflammation. A rapid reduction in anti-GBM antibody titers is necessary in view of the speed of glomerular damage, and this cannot efficiently be achieved by drug therapy alone.

Rapid initiation of treatment is critical because recovery of kidney function is much more likely in the early phase of disease before oliguria develops or dialysis is required. Albumin is the replacement fluid of choice, but fresh-frozen plasma (FFP) will be needed in patients with pulmonary hemorrhage or those who are at increased risk for bleeding. Long-term outcome of anti-GBM antibody disease treated with plasma exchange and immunosuppression was reported by Levy in an uncontrolled study, on the long-term outcomes in 71 patients with anti-GBM disease.⁴⁰ All patients received a standard immunosuppressive regimen of oral prednisolone and oral cyclophosphamide plus TPE. Plasma exchange (50 mL/kg, to a maximum of 4 L) was performed by using a centrifugal cell separator daily for at least 14 days or until anti-GBM antibody was undetectable. Human albumin (5%) with added calcium and potassium was used as replacement fluid, and FFP (150–300 mL at the end of the exchange) was used in patients with recent surgery or renal biopsy and those with pulmonary hemorrhage. Overall patient survival was 81% at 1 year of follow-up (95% in those with a creatinine level <5.7 mg/dL and 65% in those who presented with dialysis-dependent renal failure). In patients who presented with a serum creatinine concentration >5.7 mg/dL (but did not require immediate dialysis), renal survival was 82% at 1 year and 69% at last follow-up. There was only 8% renal survival in patients presenting with dialysis-dependent kidney failure, and all patients who required immediate dialysis and had 100% crescents on kidney biopsy remained dialysis dependent. The conclusion was that all patients with anti-GBM antibody disease and severe kidney failure who do not require immediate dialysis should be treated with aggressive immunosuppression and intensive TPE. Because pulmonary hemorrhage is associated with high mortality, TPE should be initiated in patients with anti-GBM disease, regardless of the severity of the kidney failure.

A retrospective review⁴¹ of 221 patients with anti-GBM disease from a single center in China comparing corticosteroids alone and corticosteroids plus cyclophosphamide with corticosteroids plus cyclophosphamide and TPE found overall 72.7% patient survival and 25% renal survival at 1 year. The combination therapy of TPE plus corticosteroids and cyclophosphamide had an overall beneficial effect on patient survival (hazard ratio [HR] for mortality, 0.31 [0.16–0.63]; $P = .001$) and renal survival (HR for renal failure, 0.60 [0.37–0.96]; $P = .032$). In 96 patients with kidney-limited disease (no hemoptysis), the risk factors for end-stage kidney disease (ESKD) were oliguria or anuria (HR, 3.34; 95% confidence interval [CI], 2.03–5.50; $P < .001$), initial serum creatinine (doubling from 1.5 mg/dL; HR, 2.13; 95% CI, 1.65–2.76; $P < .001$), and the percentage of crescents (increased by 20%; HR, 1.83; 95% CI, 1.34–2.48; $P < .001$).

Only the combination therapy of TPE plus corticosteroids and cyclophosphamide had a beneficial effect on renal survival (HR for renal failure, 0.41; 95% CI, 0.23–0.73; $P = .002$). Only 2 of 63 patients (3%) with presenting serum creatinine more than 6.8 mg/dL remained off dialysis at 1 year. Treatment with corticosteroids plus cyclophosphamide without plasmapheresis did not result in improved renal function ($P = 0.73$). Several subsequent studies have had similar findings.^{42,43} A more recent analysis found that renal survival was about 30% at 1 year.⁴⁴

To date, there has only been one small randomized trial in anti-GBM disease,⁴⁵ which compared the addition of TPE to cyclophosphamide and corticosteroids on the clinical course of 17 patients with anti-GBM disease. Only two of eight patients who received TPE became dialysis dependent compared with six of nine in the immunosuppression-alone group, suggesting a trend toward better outcomes in the TPE group. Analysis of laboratory data and pathology showed the initial serum creatinine concentration and number of crescents were associated with better outcomes rather than therapeutics used. Patients with crescents <30% and preserved renal function did well in either arm of the study.

Immunoadsorption (IAS) is another modality of extracorporeal therapy that is currently being explored. IAS may be more efficient at clearing anti-GBM antibodies than is seen with plasma exchange. However, plasma exchange removes other factors that may have roles in inflammation and coagulation that could affect the response to treatment. A small study⁴⁶ showed potential comparable outcomes of IAS to plasma exchange therapy. A newer study showed similar effectiveness for antibody reduction in patients with both anti-GBM and ANCA-associated kidney disease.⁴⁷ These studies suggest that IAS could be considered in patients as an alternate therapy, but the lack of randomized studies and the unfamiliarity with IAS in most centers indicate that more study and experience are needed.

After kidney transplantation, allograft survival rates are comparable between patients with anti-GBM disease and those with ESKD due to other causes, although up to 50% may show linear immunoglobulin G (IgG) staining of the glomerular basement membrane. Comparable rates are only in context of kidney transplantations in patients with anti-GBM disease that have taken place only after maintenance of 6 to 12 months of seronegativity. On the other hand, transplantation in the presence of antibodies leads to disease recurrence in up to 50% of allografts.⁴⁸ The immunosuppression used to prevent allograft rejection and the sustained disappearance of anti-GBM antibodies are thought to be the main reasons why recurrences in the allograft are rare.⁴⁹ Overall, long-term patient and graft survival rates are excellent.⁵⁰

RAPIDLY PROGRESSIVE GLOMERULONEPHRITIS

RPGN is characterized by rapid deterioration in kidney function occurring over a few days to a few weeks and, if untreated, can lead to ESKD. RPGN is characterized by severe inflammation and necrosis of most glomeruli and, frequently, by fibrocellular crescents (crescentic glomerulonephritis, GN). There are three major subgroups of RPGN: 1. anti-GBM disease and Goodpasture syndrome

(discussed previously); 2. immune complex-mediated processes where there is immune deposition usually resulting from autoimmune diseases such as systemic lupus erythematosus (SLE), postinfectious processes, mixed cryoglobulinemia, and IgA nephropathy; and 3. pauci-immune diseases that are usually (~80% of patients) associated with antineutrophil cytoplasmic antibodies (ANCA) including granulomatosis with polyangiitis (GPA) or microscopic polyangiitis (MPA). There are limited data for the role of TPE in immune complex-mediated processes.

The rationale for using TPE in pauci-immune ANCA-associated diseases (e.g., GPA and MPA) was initially based on the similarity of the renal pathology of these disorders with anti-GBM disease, and some patients will have both anti-GBM and ANCA-related disease. Several small case reports have documented cases of dual-positive ANCA-associated vasculitis and anti-GBM disease; however, no standardized guidelines exist for managing this overlap syndrome. The documented cases generally follow treatment protocols similar to those for anti-GBM disease, incorporating TPE, corticosteroids, and immunosuppressive therapy to target both antibody-mediated and inflammatory components of the disease. The first use of TPE for the treatment of RPGN associated with GPA was reported in 1977 when the combination of plasmapheresis, oral prednisolone, and cyclophosphamide was associated with rapid recovery of kidney function in five of nine patients.⁵¹ However, several studies throughout the 1990s did not demonstrate an additional benefit for the use of TPE in the treatment of ANCA-associated diseases. For example, Hammersmith Hospital reported a controlled trial of TPE in focal necrotizing glomerulonephritis on 48 patients randomized to conventional treatment with oral steroids, cyclophosphamide followed by azathioprine, with or without intensive TPE (at least five exchanges in the first 7 days). There was no benefit in patients with moderate or severe kidney disease who were not dialysis dependent at presentation.⁵² However, this study was the first to suggest that some patients who were dialysis dependent may be able to discontinue dialysis after treatments that included TPE (10/17 in the TPE group vs. 3/8 in the conventional group).

The Canadian Apheresis Study Group randomized 32 patients with RPGN to receive intravenous methylprednisolone, followed by oral prednisolone, and azathioprine with or without TPE (10 exchanges in the first 16 days). Again, there was no demonstrable benefit of TPE in non-dialysis-dependent patients; however, three of four dialysis-dependent patients who received TPE were able to come off dialysis compared with only two of seven patients who did not receive TPE.⁵³

In one of the largest studies, Jayne and colleagues⁵⁴ reported on 137 patients with ANCA-associated systemic vasculitis confirmed by kidney biopsy and serum creatinine concentrations higher than 5.7 mg/dL. Patients were randomized to receive seven TPE procedures ($n = 70$) or 3 g of methylprednisolone in divided doses ($n = 67$). Both groups received oral cyclophosphamide and oral prednisolone as maintenance therapy. Dialysis independence at 3 months was the primary endpoint, and 33/67 (49%) of methylprednisolone-treated patients were alive and independent of dialysis compared with 48/70 (69%) of patients who received TPE ($P = .02$). When

compared with methylprednisolone, TPE was associated with a 24% reduction in risk of progression to ESKD at 1 year.

A meta-analysis has supported the concept that TPE may reduce dialysis dependence, although there was no benefit on survival. RCTs for treating renal vasculitis were included in the analysis (31 trials), and 8 included adjuvant plasma exchange. At 12 months, there was a significant reduction in the need for dialysis in 6 trials with 235 patients (relative risk, 0.45; 95% CI, 0.29–0.72); the number needed to treat was 4 to 10.⁵⁵ However, the PEXIVAS trial randomized 704 patients with eGFR <50 mL/min or pulmonary hemorrhage to 7 TPE sessions within 14 days. All patients received cyclophosphamide or rituximab and pulse steroids. Although confidence intervals were wide, there was no benefit to receiving TPE on overall or renal survival at 1 year.⁵⁶ In conclusion, although study sizes are relatively small, presentations vary in disease severity and the provision of TPE cannot be blinded. Available data in aggregate support a beneficial role for TPE in the treatment of patients with ANCA-associated vasculitis and severe kidney disease (sCr >500 μ mol or 5.7 mg/dL) at presentation or with relapsing disease. In patients with ANCA- and anti-GBM-associated disease presenting with diffuse pulmonary alveolar hemorrhage, TPE may be beneficial for pulmonary recovery and reduction of risk of progression to dialysis.^{57–59} Therefore TPE is, at present, the best adjunct to immunosuppressive therapy for patients with advanced kidney disease from these mechanisms of injury. Nevertheless, the use of TPE for less severe kidney disease is still not clear.⁶⁰

Clinical Relevance

In patients with ANCA vasculitis, the PEXIVAS trial did not show a benefit of TPE on progression to ESRD or mortality. However, numerous smaller studies and an older systematic review suggest that TPE may reduce dialysis dependence in ANCA vasculitis. In patients presenting with diffuse alveolar hemorrhage or severe acute kidney injury, there may also be a benefit from TPE by removing toxic antibodies that cause tissue injury.

TPE reduces mortality in many acquired forms of TTP/HUS; where mortality once reached 90%, TPE has reduced this to about 10%. Although the etiologies vary, conditions where ADAMTS13 levels are <10% of normal are most likely to respond to TPE. A PLASMIC score >5 can help identify TTP earlier, enabling more rapid treatment. PLASMIC scores of ≤ 5 may still benefit from TPE, and clinical judgment based on the overall presentation is needed to make the decision.

In patients with plasma cell dyscrasias and hyperviscosity syndrome (most commonly seen with elevated immunoglobulins such as IgA or IgM), TPE lowers the total protein, reduces serum viscosity, and improves symptoms. In multiple myeloma with elevated serum free light chains, TPE can reduce the free light chain burden potentially limiting kidney damage. However, studies have not concluded a clear benefit of TPE on outcomes. Newer approaches using high-cutoff dialyzers and immunoadsorption techniques may have promise in the future.

LUPUS NEPHRITIS

Acute and chronic kidney disease (CKD) are common and potentially serious complications of SLE. Management of patients with active lupus nephritis (LN) comprises an induction treatment, followed by long-term maintenance therapy. The mainstay of LN treatment has been based on the use of corticosteroids plus cyclophosphamide (IVC) or mycophenolate in patients with active disease. Rituximab has been considered as an alternative in the treatment of proliferative LN, but data from a randomized controlled study⁶¹ suggested no additional benefits when looking at outcomes of remission with rituximab and mycophenolate versus mycophenolate alone. Newer therapies including voclosporine and belimumab have efficacy in LN, but none have been studied in combination with TPE.

The use of TPE for patients with SLE was first reported in the 1970s⁶² where removal of high levels of circulating immune complexes showed clinical improvement. However, it was not until the results of the Lupus Nephritis Collaborative Study Group in 1992 that there was a randomized study⁶³ to examine the safety and efficacy of TPE systematically. The Lupus Nephritis Collaborative Study Group was a large, randomized, controlled multicenter trial comparing a standard therapy regimen of prednisone and cyclophosphamide with a regimen of standard therapy plus TPE in patients with severe LN. In this study, 46 patients were randomized to the standard therapy group and 40 were randomized to the TPE group. Histologic categories included LN types III, IV, and V. TPE was carried out three times per week for 4 weeks, and drug therapy was standardized. The mean follow-up was 136 weeks and, although patients treated with TPE experienced more rapid reduction of antibodies to double-stranded DNA and cryoglobulins, the addition of TPE did not improve clinical outcomes. Kidney failure developed in 8 of 46 patients (17%) in the standard therapy group compared with 10 of 40 patients (25%) in the TPE group, and 6 of 46 (13%) died in the standard therapy group compared with 8 of 40 (20%) in the TPE group. Results were similar in magnitude and direction after an extended follow-up of 277 weeks.

Other small trials have had similar negative results,⁶⁴ while some studies have suggested benefit. TPE and pulse intravenous cyclophosphamide, followed by oral cyclophosphamide and prednisone, were trialed in an uncontrolled study of 14 patients with severe lupus. All 14 patients responded, and 8 remained off therapy for 5 to 6 years. One patient had a major relapse, and two others had a minor relapse at 2 and 3 years. The main adverse effect was herpes zoster; in addition, 4 of 14 women developed irreversible amenorrhea. Danieli and colleagues⁶⁵ compared two groups of patients with proliferative LN at 4 years of follow-up. Group I (12 patients) received synchronized therapy with TPE and cyclophosphamide, whereas group II (16 patients) received intermittent cycles of cyclophosphamide. At the end of the follow-up, the patients who received synchronized therapy had a faster remission than the other group, but renal outcomes were not superior to conventional therapy at long-term follow-up.

Although the current literature does not support a clear benefit for the addition of TPE to immunosuppressive therapy for LN, it may be considered for select patients with

severe disease and no response to conventional treatment. IAS may be the preferred option, compared with PE, due to safety profiles. As a result, the American Society for Apheresis does not consider TPE for LN but TPE should be considered in patients with catastrophic antiphospholipid antibody syndrome in patients with lupus or refractory severe disease (category II; 2C).

MIXED CRYOGLOBULINEMIA

The first time that cryoglobulinemia was identified in the blood of a patient with multiple myeloma, where it was noted that cooling of the patient's blood resulted in a precipitate forming which disappeared upon subsequent heating, was 1933. Cryoglobulinemia refers to the presence of serum proteins that precipitate at temperatures below 37°C and redissolve on rewarming. When purified, they can be categorized into three main groups: monoclonal (type I cryoglobulinemia), mixed monoclonal and polyclonal (type II cryoglobulinemia), and polyclonal IgG and IgM (type III cryoglobulinemia). The immunochemical composition of cryoglobulins often correlates phenotypically with clinical symptomatology of disease. More than 80% of patients with mixed cryoglobulinemia are infected with hepatitis C virus (HCV),⁶⁶ although cryoglobulinemia can also be seen in plasma cell dyscrasias and with vasculitis. The glomerular injury is the consequence of glomerular deposition of immune complexes, and the kidney's manifestations may range from isolated proteinuria to overt nephritic or nephrotic syndrome, with variable progression toward ESKD.^{41,42} There have been no randomized, controlled studies of TPE for cryoglobulinemia; however, the removal of pathogenic cryoglobulins is rational, and there have been numerous anecdotal case reports and uncontrolled studies showing that TPE may benefit patients with severe clinical manifestations of cryoglobulinemia such as progressive kidney failure, severe or malignant hypertension, purpura, and advanced neuropathy.^{43,44} Uncontrolled studies with more than five patients have shown that TPE induces rapid reduction in the "cryocrit" (the proportion of plasma comprised of cryoglobulin), improved kidney function in 55% to 87% of patients, and improved survival (~25% mortality rate) compared with historical data (~55% mortality rate).⁴⁵ Plasma exchange does not prevent the formation of new cryoglobulins or treat the underlying disease, and rebound production can occur after cessation of TPE without therapy directed at the cryoglobulin-producing B cell clones or treatment of the underlying disease process. The unique characteristics of cryoglobulins have led to modifications of the TPE technique to enhance their removal. Cryofiltration is a modified technique that reduces circulating levels of cryoproteins. In this technique, the plasma is cooled in the extracorporeal circuit to precipitate the proteins, allowing a more efficient removal of the pathogenic proteins.⁶⁷

In a multicenter retrospective study⁶⁸ that included 159 patients with cryoglobulinemia, including 113 HCV-positive patients, it was shown that the TPE led to complete remission in 12% of cases, partial remission in 38% of cases, transient remission in 25% of cases, and no remission in 22% of cases. The treatment of hepatitis C has been revolutionized with the introduction of direct antiviral agents, but few data

are available on the safety and efficacy of antiviral therapy with direct-acting antivirals in the treatment of HCV-related cryoglobulinemia. Some trials have shown that patients treated with sofosbuvir-based, direct-acting antiviral therapy experience improvement in kidney function.^{68a}

Although therapies for type I and type II cryoglobulinemia (such as immunosuppression, antivirals, or chemotherapy) are quite effective, they take time to work. Given that cryoglobulins can affect multiple organs, use of different modalities of apheresis can be an effective way to rapidly decrease these serum proteins, providing time for other therapies to take effect.

The American Society of Apheresis classified severe/symptomatic cryoglobulinemia a category 2, meaning it is reasonable to use apheresis as a second-line therapy using either plasmapheresis (grade 2A) or immunoadsorption (IA; grade 2B).

KIDNEY DISEASE ASSOCIATED WITH MULTIPLE MYELOMA AND OTHER HEMATOLOGIC DISORDERS

Kidney injury is a common finding in MM, affecting 20% to 50% of patients at diagnosis, and 10% to 15% will develop ESKD. Kidney injury can be caused by a variety of factors, including precipitation of myeloma light chains in renal tubules (Bence Jones proteins) that can lead to direct tubular toxicity (cast nephropathy). In addition, paraproteins can cause numerous other pathologic patterns of injury, including MPGN, C3 nephropathy, and light and heavy chain deposition disease and can be associated with rare forms of glomerulonephritis such as fibrillary and immunotactoid glomerulonephritis. Other factors frequently implicated in MM-associated kidney failure include hypercalcemia, hyperuricemia, amyloidosis, hyperviscosity, infections, and chemotherapeutic agents.⁶⁹ A kidney biopsy is needed to establish the mechanism of injury but may not be obtained if the MM is being treated. This has confounded studies on the role of TPE.

Given the high association of kidney disease in MM, it is important to identify the diagnosis early to improve renal outcomes (CKD and ESRD). Before use of modern chemotherapeutic methods, patients with AKI and MM on dialysis had <25% chance of renal recovery.⁷⁰ Currently, the combination of the proteasome inhibitor bortezomib with dexamethasone is the standard regimen in newly diagnosed patients and has shown to suppress light chain production. The rationale to use TPE in MM is with the goal to decrease light chains levels quickly (as the literature shows lower light chain levels are associated with improved renal outcomes and survival). In 2018, Premuzic and colleagues⁷¹ published a study of 29 patients with MM and AKI who underwent treatment with bortezomib or bortezomib and plasmapheresis; the pheresis group showed a more pronounced significant reduction in light chain level (serum free light chains decreased from 2422 mg/L to 34 mg/L in the TPE arm vs. a decrease from 2428 mg/L to 182 mg/L in the bortezomib-alone arm). There was also a correlation of better outcomes with higher number of sessions of TPE. However, there have been a few RCTs⁷² where TPE was not shown to reduce adverse renal outcomes or death. However, this was a relatively small study of 104 patients with AKI at the onset of myeloma. Importantly,

a kidney biopsy was not needed for inclusion, so it is unclear whether a subset of patients had benefit.

TPE only removes 25% of light chain volume in 3 weeks, so this has prompted an exploration of newer methods of light chain removal by modifying aspects of the hemodialysis procedure.^{72a} Most (90% of) light chains could be removed by high-cutoff dialysis membranes (HCO-HD) with extended dialysis times in 3 weeks. Changing dialyzers during extended HD and increasing UF rates were also shown to be associated with light chain removal rates.⁷³ Newer studies have shown increased light chain removal with the use of HCO-HD.^{74,70} Most of these studies were retrospective and adjustments for baseline eGFR status, percentage of plasma cells, renal biopsy findings, etc. were not consistently reported.

A small retrospective study⁷⁵ looked at AKI, specifically in MM, and compared the use of chemotherapy alone versus chemotherapy along with CVVH and hemoperfusion. After 1-year follow-up, the improvement in renal function was significant in the chemotherapy with CVVH and hemoperfusion combination group. This group had 14 complete remissions and 19 partial remissions compared with 9 complete remissions and 11 partial remissions in the chemotherapy-alone group, a statistically significant result. Another approach has been to use multiple polymethylmethacrylate membrane filters in the course of the dialysis treatment. This is based on the observation that free light chains bind to and saturate the membrane filter. In one study⁷⁶ on patients presenting with AKI and MM requiring HD, the use of two polymethylmethacrylate membranes along with bortezomib allowed reduction of FLC to <200 mg/L. Close to 70% of patients showed renal recovery in 60 days.

In aggregate, these studies leave the role of TPE and modified dialysis techniques in the management of myeloma-associated kidney injury unresolved. Questions remain about subgroups of patients who may benefit, and individuals should be assessed on a case-by-case basis while larger studies are needed.

Waldenström macroglobulinemia (WM) is a B cell disorder resulting from the accumulation of clonally related IgM-secreting lymphoplasmacytic cells. The morbidity associated with WM is typically mediated by tissue infiltration by neoplastic cells and by the physicochemical and immunologic properties of the monoclonal IgM. In patients with symptomatic hyperviscosity, cryoglobulinemia, or moderate to severe cytopenias, emphasis should be placed on achieving rapid reduction in the burden of plasma paraproteins. This was demonstrated in a study where 95 patients with symptomatic hyperviscosity (81%) received plasma exchange as the initial therapy, while 23 were treated with chemotherapy alone. Following plasma exchange, 90% of patients experienced a resolution of hyperviscosity-related symptoms. The median reduction in serum IgM and serum viscosity after plasma exchange during the first episode was 2860 mg/dL, with a range from 260 to 9900 mg/dL. In these circumstances, TPE of one or two sessions was effective in reducing serum IgM levels.⁷⁷ Treatment should be initiated as soon as possible, with a regimen including bortezomib, dexamethasone, and rituximab to achieve more rapid disease control.⁷⁸

TPE has been widely used in hematologic and oncologic diseases; however, only the following disorders are

considered category I by the American Society for Apheresis: 1. TTP (discussed later); 2. polycythemia vera with severe erythrocytosis; 3. sickle cell disease (red cell exchange); 4. hyperviscosity in hypergammaglobulinemias; and 5. cutaneous T-cell lymphoma (photopheresis).¹

SYNDROMES OF THROMBOTIC MICROANGIOPATHY

The thrombotic microangiopathies (TMAs) are multisystem disorders that can affect children and adults, may be hereditary or acquired, and share a spectrum of abnormalities in numerous organ systems. The hallmark clinical features of this syndrome include microangiopathic hemolytic anemia, thrombocytopenia, and organ (including kidney) injury. The pathologic features are vascular damage manifested by arteriolar and capillary thrombosis. Common causes of this syndrome include toxins from bacteria or drugs, abnormal complement activation that can be genetic or autoantibody-induced, production of procoagulant factors (antiphospholipid antibodies), loss of anticoagulants (defects on ADAMTS13), and severe hypertension.^{79,80} According to the latest ASFA guidelines, the TMA diseases with class I recommendation for TPE procedures are thrombotic TTP, complement-mediated TMA when it is associated with factor H antibodies, and ticlopidine-associated TMA.¹

TTP, also known as TMA-ADAMTS13 deficiency, is a systemic thrombotic illness affecting mostly small vessels. Presenting clinical features of acquired TTP are weakness, gastrointestinal symptoms, purpura, and transient focal neurologic abnormalities. Clinical presentation is diverse because some patients have minimal abnormalities, whereas others are critically ill. Diagnostic criteria are the presence of microangiopathic hemolytic anemia and thrombocytopenia without another apparent cause; an ADAMTS13 level of <10% of normal activity supports the clinical diagnosis of acquired TTP.^{81,82} Multimers of von Willebrand factor (VWF) normally accumulate on the endothelial cell membrane and are rapidly cleaved into normal-sized multimers by the ADAMTS13 protease. TTP results from the accumulation of unusually large VWF (ULVWF) due to insufficient ADAMTS13 activity. The accumulation of ULVWF multimers results in platelet microthrombi formation and subsequent microangiopathic hemolytic anemia. An inhibitory autoantibody to the ADAMTS13 metalloproteinase has been found at varying titers among a high percentage of patients with the idiopathic form of this disease.⁸³

Before the introduction of plasma infusion and TPE, the disease rapidly progressed and was almost uniformly fatal (90% mortality). In 1977, it was discovered that infusion of FFP or TPE with FFP replacement was able to reverse the course of disease.^{84,85} The PLASMIC score is a clinical tool used to assess the probability of TTP in patients with TMA while awaiting ADAMTS13 test results.⁸⁶ It consists of seven components, and patients are awarded 1 point for each laboratory parameter, including the following:

- Platelet count (<30,000/mm³)
- Hemolysis evidence (indirect bilirubin >2 mg/dL or LDH >2× ULN)
- Absence of active cancer (1 point if no active cancer)

- No solid organ or stem cell transplant history (1 point if no such history)
- Mean corpuscular volume (MCV) (<90 fL)
- International normalized ratio (INR) (<1.5)
- Creatinine level (<2 mg/dL)

A review and meta-analysis⁸⁷ looked at how well the PLASMIC score predicts TTP in patients, using data from 13 studies with 970 patients. TTP was present in 35% of cases. A PLASMIC score of 5 or higher had a sensitivity of 99% and a specificity of 57%. Scores of 6 or higher lowered sensitivity to 85% but increased specificity to 89%. These results suggest that patients with a score of 5 or more should be treated for TTP unless another clear diagnosis explains their symptoms without waiting for confirmation of the ADAMTS13 assay.

Currently, daily TPE is the mainstay treatment by removing the inhibitory autoantibody and repletion ADAMTS13 with FFP. Plasma exchange with FFP is more effective than plasma infusion alone; at 6 months, the remission rate is 78% versus 31%, and the survival rate with these two procedures is 78% versus 50%.⁸⁸ If there is a delay in initiating TPE, large-volume plasma infusions are indicated as initial therapy.⁸⁹ The optimal duration of TPE is not known but, in TTP, TPE is performed daily until the platelet count has risen to near-normal and evidence for hemolysis (schistocytes, lactate dehydrogenase [LDH] elevation) has resolved. There is a wide range of reported exchanges (3–145), with an average of 7 to 16 daily exchanges necessary to induce remission.^{82,90} The ASFA guidelines recommend daily TPE until the platelet count is above 150,000/L for 2 or 3 consecutive days and the LDH level is close to normal.⁹¹

When present, neurologic symptoms usually improve rapidly, and the serum LDH level tends to improve over the first 1 to 3 days. Improvement in the platelet count may not be seen for several days, however, and improvements in kidney function often take longer. Patients requiring dialysis at presentation may be able to recover enough function to discontinue dialysis, but many patients have residual CKD. When a normal platelet count has been achieved, plasma exchange is gradually tapered by increasing the interval between treatments. Many patients (one-third to one-half) will abruptly develop recurrent thrombocytopenia and increased evidence of hemolysis when daily plasma exchanges are tapered or stopped. Some of these patients may benefit from the addition of prednisone or other immunosuppressive therapy (e.g., cyclosporine and rituximab), although there are sparse data evaluating the safety, efficacy, and effectiveness of these agents.

Although there is no evidence for a beneficial role of TPE in patients with hemolytic uremic syndrome (HUS), TTP is often secondary to drugs, including cancer chemotherapy and calcineurin inhibitors or when associated with bone marrow transplantation.⁹² There is a single uncontrolled report of 60 patients with ticlopidine-associated TMA, suggesting that TPE enhances survival (50% vs. 24% mortality).⁹³ One well-defined cause of HUS (now called TMA–Shiga toxin-mediated) is the syndrome associated with hemorrhagic diarrhea caused by *Escherichia coli* O157:H7. In this disease, the enterotoxin induces colonic vascular injury, leading to systemic absorption and activation of numerous pathways, causing endothelial

cell damage over several days. Platelet microthrombi are particularly prominent in the glomerular capillaries and often lead to severe AKI. Supportive care with fluid resuscitation and timely renal replacement therapy is the mainstay of therapy, glucocorticoids are not indicated, and there is no compelling evidence from the literature that TPE benefits patients with this syndrome. However, patients with severe bloody diarrhea or neurologic involvement may respond to TPE. During the 2011 outbreak in Germany, TPE was carried out in 87% patients, yet evidence of benefit was not seen, and improvement in general medical care rather than the frequent and early use of TPE was the potential explanation of the better outcome in comparison with historic controls. A group of patients with more severe disease received TPE plus eculizumab, but survival was not enhanced with addition of this agent.⁹⁴

KIDNEY TRANSPLANTATION

TPE has been employed in a variety of clinical settings related to kidney transplantation. These include recurrent primary focal glomerulosclerosis (FSGS) in transplanted kidneys, ABO blood group-incompatible transplants, positive T-cell crossmatch, and acute humoral rejection.

RECURRENT FOCAL SEGMENTAL GLOMERULOSCLEROSIS

Primary FSGS is attributed to the cause of end-stage renal failure in approximately 2.3% of ESRD patients⁹⁵ and primary FSGS reoccurs in approximately 30% in kidney transplant recipients.⁹⁶ In addition, the risk of relapse is higher (80%–90%) in patients with a prior history of allograft loss due to recurrent FSGS.⁹⁷ Additional factors associated with an increased risk of recurrence are rapid progression to ESKD, unremitting proteinuria, mesangial hypercellularity, and cellular variant.⁹⁸

The mechanisms of recurrent FSGS are unclear, but the early reappearance of proteinuria suggests the presence of a nondialyzable circulating factor that alters glomerular permeability. Removal of a circulating factor by IA or plasma exchange may account for the remission of the disease in some patients. To date, several circulating factors have been proposed to be the triggers of FSGS, including the soluble urokinase plasminogen activating receptor (suPAR), anti-CD40 autoantibody, and cardiotrophin-like cytokine factor-1 (CLCF1). suPAR is a soluble (circulating) form of an otherwise membrane-bound, three-domain receptor for urokinase that mediates extracellular proteolysis. The suPAR signaling protein is expressed on a variety of cells, including leukocytes, endothelial cells, podocytes, and immature myeloid cells. suPAR has been heavily studied in connection with FSGS, particularly because of its role as a potential biomarker and pathogenetic factor in kidney disease.⁹⁹ Whether suPAR is a direct cause of FSGS or merely a byproduct of systemic inflammation remains conflicting.¹⁰⁰

Treatment options include immunosuppressive drugs (e.g., cyclophosphamide, methylprednisolone, high-dose cyclosporine, and rituximab) and TPE. TPE has been used as a first-line treatment for recurrent FSGS, but there are few randomized controlled studies, and most reports have been retrospective reviews of a single

center, often without controls. Nevertheless, TPE has been widely applied in the prevention and treatment of recurrent FSGS since the initial 1985 report by Zimmerman.¹⁰¹ Since then, numerous systematic reviews have suggested that TPE promotes partial or complete remission in patients with recurrence of prior FSGS. In 2005, Gohh and colleagues¹⁰² examined a cohort of 10 kidney transplant patients with recurrent FSGS and found that plasmapheresis, particularly when combined with immunosuppressive agents, was able to induce partial or complete remission in 7 of them. Ofatumumab¹⁰³ or abatacept¹⁰⁴ has been used an adjunct in disease refractory to TPE and rituximab.

Preemptive plasmapheresis in renal transplantation for FSGS is a proactive approach used to prevent the recurrence of FSGS in high-risk patients (such as those with a history of recurrent FSGS in a previous transplant or those with aggressive disease). Ohta and colleagues¹⁰⁵ have reported the recurrence rate of primary FSGS in patients who received preoperative TPE. TPE was performed in 15 patients and recurrence developed in 5 (33%), whereas 4 of 6 (66%) without preoperative TPE developed recurrence. In a multicenter cohort study involving 75 patients who experienced recurrence of FSGS after kidney transplantation, 61 patients were treated with TPE, with or without rituximab. Among these patients, complete or partial remission was achieved in 35 individuals (57%), while 26 patients (43%) showed no response to the treatment.⁹⁶

An alternative approach has been to use IA columns. IA columns are designed to selectively bind and remove harmful immunoglobulins and circulating proteins (such as the suspected permeability factor in FSGS) from the blood. Unlike plasmapheresis, which nonspecifically removes all plasma components, IA is more targeted and less likely to deplete beneficial proteins.¹⁰⁶ IA's effectiveness suggests that the circulating factor in FSGS might be related to immunoglobulins or reside within the immunoglobulin fraction. Veissi¹⁰⁷ sought to purify immunoglobulin fractions from the plasma of FSGS patients, presumed to carry a circulating factor, and assess their impact on human conditionally immortalized podocytes (hCiPods) in vitro. The results showed that the suspected circulating factor associated with FSGS was not present in the immunoglobulin fractions from the limited three-patient cohort examined. These findings imply that IA alone may not be sufficient to effectively eliminate circulating factors or that the circulating factor is not an immunoglobulin. IA is a promising therapeutic option for treating recurrent FSGS in kidney transplant recipients. Its targeted approach offers advantages over plasmapheresis, including fewer adverse effects and potentially better outcomes in certain patients. However, IA's limited availability and high costs indicate further research is needed to fully establish its efficacy and role in the management of FSGS.

In summary, plasmapheresis remains a cornerstone in the management of recurrent FSGS after renal transplantation, although its success is variable. It is most effective when combined with other immunosuppressive therapies and initiated early. Ongoing research into the exact mechanisms of FSGS recurrence and the identification of circulating factors may lead to more targeted treatments and better outcomes.

ABO-INCOMPATIBLE KIDNEY TRANSPLANTATION

The ABO blood group antigen system was discovered on red blood cells by Landsteiner in 1901; these antigens are expressed throughout the body and, in the kidney, are found in the distal tubules, collecting tubules, and vascular endothelium of peritubular and glomerular capillaries. The ABO antibodies (isoagglutinins) are produced in the first years of life by sensitization to environmental substances such as food, bacteria, and viruses and are usually of the IgM type. These antibodies against ABO antigens generally prohibit kidney transplantation across ABO barriers and are key mediators of antibody-mediated rejection.

In the early days of kidney transplantation, results with ABO-incompatible (ABOi) transplants were disappointing, but a major breakthrough came with the first large study on ABOi kidney transplantation reported in the Netherlands.¹⁰⁸ Recipients received two to five TPE sessions, pretransplantation immunosuppression, and splenectomy. Graft survival rates were better in younger patients (89% at 5 years in patients <15 years old compared with 77% in older patients). In Japan, ABOi renal transplantation flourished in the 1990s and the outcomes to date have been excellent. In the first report of 441 ABOi kidney transplantations performed at 55 centers across Japan from 1989 to 2001, graft survival was 84% in the first year and 59% at 9 years of follow-up. Allograft survival was not significantly lower than in historical controls who received ABO-compatible living donor organs. For the period of 2001 to 2010, patient and graft survival rates for the 1427 analyzed patients were an excellent 98% and 96% for the first year and 83% and 91% after 9 years, respectively.¹⁰⁹ The protocol to prepare recipients consisted of four components: 1. extracorporeal immunomodulation to remove AB antibodies before the transplantation; 2. use of immunosuppressive drugs; 3. splenectomy; and 4. anticoagulation. TPE and IA were the two techniques performed to remove the AB antibodies, and the goal with either technique was to decrease pretransplantation serum AB titers by 8- to 16-fold. Antibody removal was usually not performed after transplantation. Splenectomy has also been replaced by the anti-CD20 antibody rituximab.

The Johns Hopkins group has established a preconditioning protocol of TPE, cytomegalovirus (CMV) hyperimmune globulin (CMVlg), and anti-CD20 (rituximab) to allow ABOi kidney transplantation without splenectomy. The treatment protocol requires four to five preoperative TPE sessions to remove anti-A/B antibodies, and each session is followed by the administration of CMVlg. After achieving a pretransplant A/B-antibody titer <1:16, a single dose of rituximab is given 1 or 2 days before transplantation. Thereafter, immunosuppression with tacrolimus and mycophenolate mofetil is initiated, followed by steroids and daclizumab after transplantation. Postoperative treatment included another three TPE-CMVlg sessions on days 1, 3, and 5. The 5-year graft survival rate for the cohort of 60 consecutive patients was 88.7%.¹¹⁰ Data from the Scientific Registry of Transplant Recipients on the outcomes of 738 ABOi kidney transplantations performed between 1995 and 2010 in the United States have shown a cumulative incidence of graft loss during the first year of 5.9% in

ABOi (vs. 2.9% in ABO-compatible recipients). This occurred mainly during the first 2 weeks after surgery due to rejection.¹¹¹ ABO-incompatible (ABOi) transplant strategies have largely disappeared from clinical practice at most centers. However, for patients with Anti-A or Anti-B antibodies, where isoagglutinin titers at the time of transplantation are closely linked to the risk of early antibody-mediated rejection, apheresis is effective in lowering anti-A and/or anti-B antibody levels. While ideal postdesensitization anti-A/B titers have not been precisely defined, they typically range between 1:8 and 1:32. This reduction helps to minimize the risk of rejection during ABOi transplants, but the exact threshold for optimal outcomes remains under investigation. In another study, Barnett¹¹² reported patient survival rates of 94.5% at 1 year and 91.9% at 3 years in 62 ABOi kidney transplants. In comparison, patient survival rates for 167 ABO-compatible transplants were 99.3% at 1 year and 98.3% at 3 years, indicative of comparable outcomes.

Selective plasma exchange (SePE) is a specialized form of plasma exchange that utilizes a membrane plasma separator with smaller pores compared with standard separators. This allows SePE to selectively remove small- and medium-molecular-weight substances while leaving larger molecules, such as coagulation factors, intact.¹¹³ This selective filtration makes SePE useful for conditions where the goal is to reduce specific substances without affecting important larger proteins in the blood. A 2020 study¹¹⁴ investigated the use of SePE in ABOi kidney transplantation, comparing outcomes based on the substitution of fluids. The study evaluated the effects of using albumin (Group A, $n = 41$) versus a combination of albumin and FFP (Group F, $n = 17$) following immunosuppressive therapy and SePE. The comparison aimed to assess the efficacy and safety of these substitution methods in managing antibody levels and reducing the risk of rejection in ABOi transplants. The use of FFP in SePE effectively reduced isoagglutinin titers (significant decrease in Group F with lowering of IgG isoagglutinin ($P < 0.0001$) and IgM isoagglutinin ($P = 0.0044$) unlike albumin alone, which in some cases failed to lower these antibody levels (30% of Group A, titers did not decrease). SePE with FFP not only decreases isoagglutinin titers but also replenishes essential immunoglobulins and coagulation factors. As a result, this approach may offer a more advantageous treatment option as a preparatory apheresis method before ABO-incompatible kidney transplantation.

POSITIVE T-CELL FLOW CROSSMATCH

High sensitization to human leukocyte antigen (HLA) indicates positive T-cell flow crossmatches against multiple potential donors. The degree of sensitization is quantified as the percentage of the donor pool against which the serum of the patient has positive T-cell flow crossmatches—the calculated panel-reactive antibody (cPRA) status. Patients with a cPRA persistently higher than 50% are generally considered highly sensitized; primary sensitization results from exposure to foreign HLA antigens via transplantation, transfusion, or pregnancy, although infection and other conditions can also alter sensitization status. Patients with preformed antibodies against HLA antigens have a lower

probability of receiving a matched kidney from a deceased or living donor. Furthermore, presensitized recipients enjoy less favorable outcomes after deceased donor kidney transplantation and are at increased risk for hyperacute or acute antibody-mediated rejection and graft loss. Successful transplantation in these patients requires a desensitization protocol of the recipient to a specific donor to reduce the risk of hyperacute rejection and immediate graft loss.^{115,116}

The general approach in HLA antibody reduction protocols involves the use of low-dose IgG and TPE. TPE is used to remove anti-HLA antibodies and is followed by infusion of low doses of IgG during hemodialysis. The rationale is that low-dose IgG will have beneficial immunomodulating effects. Concurrent with TPE initiation, patients are treated with tacrolimus, mycophenolate mofetil, steroids, and antimicrobial prophylaxis. TPE is continued three times weekly until the T-cell flow crossmatch is negative, and transplantation usually takes place within 24 hours. TPE and low-dose IgG are usually repeated several times during the first 2 post-transplantation weeks to remove any rebounding antibody.¹¹⁷ TPE-based protocols are usually not suitable for highly sensitized patients awaiting deceased donor transplantation because the availability of suitable organs is unpredictable, and TPE is both difficult and expensive to continue indefinitely. If TPE is stopped, anti-HLA antibody titers will rebound.¹¹⁸

ACUTE HUMORAL REJECTION

Acute humoral rejection (AHR) is characterized by a severe allograft dysfunction associated with the presence of circulating donor-specific antibodies (DSAs). The primary aims of therapeutic modalities for AHR are to remove existing antibodies and inhibit their redevelopment. Poor outcomes are seen with AHR, and treatment with pulse steroids and antilymphocyte therapy is often ineffective. There are few randomized controlled studies using TPE, and most of them were conducted before current high-potency immunosuppression regimens were available. Removal of the DSAs with TPE has been successful when combined with tacrolimus and mofetil mycophenolate.¹¹⁹ Another compared the outcomes of 12 patients treated with a combination of TPE, IVIg, and rituximab, against 12 patients treated with IVIg alone. The results showed better graft survival at 36 months for the group receiving TPE and rituximab (91.7% vs. 50%).¹²⁰ A prospective observational study evaluated the role of TPE in antibody-mediated rejection (AHR) among a cohort of 1608 patients who underwent living-related renal transplantation.¹²¹ Of these, 49 patients were diagnosed with biopsy-proven AHR. The treatment protocol for AHR included TPE combined with triple immunosuppression (mycophenolate mofetil, tacrolimus, and prednisone), followed by intravenous immunoglobulin (IVIg) after TPE. If no response was observed, rituximab was added to the regimen. With this protocol, 38 out of 49 patients (78%) showed a favorable response, defined as improved urine output and at least a 50% reduction in serum creatinine. Graft survival rates were 89% at 6 months and 81% at 12 months. Among the 11 patients (22%) who did not respond to treatment, 7 required dialysis, 1 underwent graft nephrectomy followed by dialysis, and

3 patients died. The study had several limitations. DSAs were not measured before or after transplantation, which could have provided crucial insights. Additionally, 35% of patients with AHR had concurrent acute cellular rejection, potentially confounding the outcomes.

In some cases, IA has led to improved graft function or the stabilization of graft deterioration in patients with AHR. One study¹¹⁷ reported that IA combined with immunosuppressive therapies improved both short-term and long-term graft outcomes in patients with AMR. IA allows more specific and effective clearance of circulating immunoglobulins, and some authors have proposed that this technique may provide a more effective and selective antibody depletion. In general, presently in the United States, few programs are doing desensitization outside of clinical trials and instead relying on paired donation programs preferentially for these pairs.

OTHER KIDNEY DISEASES

Dialysis-related amyloidosis (DRA) is a serious complication of long-term hemodialysis, and it results from the long-term accumulation of β_2 -microglobulin in tissues. A β_2 -microglobulin-adsorbent column that connects to a dialyzer in tandem for direct hemoperfusion for DRA is commercially available. It has been most widely utilized in Japan since 1996, where there is a large cohort of long-term hemodialysis patients (>10 years). One small RCT showed improved activities of daily living after 1 year of treatment,¹²² and a larger observational study of quality of life showed better symptoms of DRA in those treated with β_2 -microglobulin columns.¹²³

Nephrogenic systemic fibrosis (NSF) is a rare but severe systemic disorder that has been rarely observed in patients with acute or CKD (more commonly in ESKD), yet it is strongly associated with exposure to select gadolinium-based contrast agents using in magnetic resonance imaging (MRI). Fortunately, the incidence of NSF has declined sharply since 2013. Case reports were published, with moderate treatment success using TPE when symptoms progressed (Category III)^{124,125}; photopheresis has also been used, with some reported success.

CATASTROPHIC ANTIPHOSPHOLIPID SYNDROME

Catastrophic antiphospholipid syndrome (CAPS) is a rapidly progressive and life-threatening disease that results in thromboses in multiple organs in the presence of antiphospholipid antibodies. Rapid-onset thromboses in multiple organs and extensive small and medium vessel involvement in atypical locations are the general characteristics of CAPS. Treatment with anticoagulation, corticosteroids, and TPE or IVIg can be initiated. TPE can remove pathologic antiphospholipid antibodies as well as cytokines, tumor necrosis factor- α (TNF- α), and complement products. Although TPE improves outcomes in patients with CAPS, note that most reported CAPS patients received TPE FFP as the replacement fluid. FFP contains natural anticoagulants (e.g., antithrombin III and protein C), as well as clotting factors, so it is unknown if TPE per se or the FFP replacement provides the benefits to patients with CAPS. There are currently no RCTs evaluating the

safety and efficacy/effectiveness of TPE in CAPS (Category I).^{126,127}

THERAPEUTIC PLASMA EXCHANGE AND NONKIDNEY DISEASE

According to several national registries, TTP, myasthenia gravis (MG), chronic inflammatory demyelinating polyneuropathy, WM, and Guillain-Barré syndrome (GBS) are the most frequent indications for plasmapheresis, and there are RCTs indicating benefits for these disorders.¹²⁸ There are now nearly 100 rational indications for TPE, and an exhaustive review of the experimental data supporting the different indications for TPE has been published by the ASFA.⁹¹ In many clinical settings, it is the nephrologist who is called for initiation of TPE; therefore it is essential that nephrologists be generally familiar with the literature supporting the use of this procedure in nonkidney diseases.

Plasma exchange is a well-established therapeutic procedure commonly used in many neurologic disorders of autoimmune cause. It is thought that the beneficial effects of TPE occur through the removal of inflammatory mediators, including autoantibodies, complement components, and cytokines. GBS, MG, chronic inflammatory demyelinating polyneuropathy, and demyelinating polyneuropathy with IgG/IgA are considered as indication category I by the ASFA.^{91,129}

GUILLAIN-BARRÉ SYNDROME

GBS has emerged in the past 25 years as the most frequently occurring clinical paralytic disorder, with an annual incidence of 2 cases/100,000 persons. GBS is dangerous, with 10% to 23% of patients requiring mechanical ventilation; up to 1 in 20 patients may die from complications of the disease. In about 60% of cases, GBS develops shortly after an infection, most commonly caused by *Campylobacter jejuni*. A large number of diverse antibodies against different glycolipids, including GM1, GD1a, and GQ1b have been described.¹³⁰ The disorder is characterized by symmetric weakness, paralysis, and distal paresthesias that usually start with the lower limbs, followed by rapid progression to the proximal limbs and trunk over several days. A typical diagnostic feature is the finding of an increased concentration of protein in the absence of a pleocytosis (albumin-cytologic dissociation) in the cerebrospinal fluid (CSF).¹³¹

TPE is a well-established treatment for GBS. Its benefits over and above supportive care have been demonstrated in two large randomized, controlled, nonblinded, multicenter trials.^{132,133} Substantial benefit has been documented for the time to recover the ability to ambulate with assistance, reduction of the proportion of patients who needed assisted mechanical ventilation, a more rapid recovery of motor function, and time to walk with and without assistance. The French Cooperative Group on Plasma Exchange in Guillain-Barré Syndrome¹³⁴ has addressed the optimal number of TPE sessions for the treatment of GBS and established that two sessions for patients with mild disability and four sessions for patients with moderate and severe disability

were the optimal number of sessions. TPE is considered equally efficacious to IVIg therapy; combined treatment of TPE and IVIg does not seem to provide an additional benefit (Category I).¹³³

CHRONIC INFLAMMATORY DEMYELINATING POLYNEUROPATHY

Chronic inflammatory demyelinating polyneuropathy (CIDP) is a common and potentially treatable disease, with an estimated prevalence of about 1 to 2 cases/100,000 adults. The occurrence of symmetric weakness in both proximal and distal muscles that progressively increases for more than 2 months is the pivotal symptom in the diagnosis of this disease. CIDP is associated with impaired sensation, absent or diminished tendon reflexes, an elevated protein level in CSF, demyelinating nerve conduction studies, and signs of demyelination in nerve biopsy specimens. The presence of autoantibodies against various proteins and glycolipids of the peripheral nerve in samples of serum and CSF from patients with CIDP may provide a rationale for the therapeutic use of TPE (Category I). The most widely used treatments for CIDP consist of IVIg, TPE, and corticosteroids. Therapy should be initiated early in the course of the disease to prevent continued demyelination and secondary axonal loss leading to permanent disability. According to published data, there appears to be no difference in efficacy among these three main therapies.^{135,136}

MYASTHENIA GRAVIS

Myasthenia gravis (MG) is an autoimmune-mediated disorder of the neuromuscular junction, clinically characterized by fluctuating muscle weakness and fatigability. The most common variant of the disease is mediated by circulating autoantibodies against the nicotinic acetylcholine receptor (AChR). Mechanisms responsible for the loss of functional AChR that compromise neuromuscular transmission include the degradation of the AChR, complement-mediated lysis of the AChR, and interference with neurotransmitter binding. In subgroups of patients negative for AChR antibody, antibodies against the receptor tyrosine kinase can be detected.¹³⁷

Treatment of MG includes thymectomy, acetylcholine esterase inhibitors, corticosteroids, immunosuppressive agents, TPE, and IVIg. It is presumed that elimination of circulating AChR antibodies and other humoral factors accounts for the observed beneficial effects of TPE. TPE is indicated where there is a need for rapid clinical improvement, including myasthenic crisis, impending crisis, preoperative stabilization, and patients in whom long-term control of symptoms is suboptimal with other forms of therapy. Occasionally, patients require long-term outpatient exchange to achieve adequate control of MG symptoms. Treatment consists of four to six exchanges, each removing 3 to 5 L of plasma, performed daily or every other day. With this TPE regimen, MG symptoms improved within 2 to 3 weeks in 65% of patients but rarely lasted more than 10 weeks without other immunosuppressive therapy. Some patients may have a more prolonged response (Category I).¹³⁸⁻¹⁴⁰

FAMILIAL HYPERCHOLESTEROLEMIA

For familial hypercholesterolemia (FH), the successful use of an apheresis procedure was first described in 1975. Clinical practice recommendations include lipid apheresis as a cornerstone in therapy for lipid lowering in homozygous FH and severe heterozygous FH when traditional lipid-lowering drugs are not sufficiently effective. It is now recommended that low-density lipoprotein (LDL) apheresis should be the treatment of choice for FH homozygotes from the age of 7 years unless their serum cholesterol level can be reduced by more than 50% or decreased to <9 mmol/L by drug therapy. It is also indicated for individual patients with heterozygous FH or a family history of premature cardiac death, progressive coronary disease, and where LDL cholesterol remains higher than 5.0 mmol/L or is decreased $<40\%$ with maximal drug therapy. Apheresis may also occasionally be indicated in patients with aggressive progressing coronary disease and lipoprotein (a)—Lp(a)—and LDL cholesterol higher than 60 mg/L and 3.2 mmol/L, respectively, despite maximal drug therapy.^{141,142} The use of PCSK9 inhibitors may obviate the role of LDL pheresis in these patients; 11 patients were successfully transitioned to biweekly evolocumab.¹⁴³

DRUGS AND PLASMAPHERESIS

TPE is effective for elimination of some toxins, depending on factors such as the toxin's clearance rate, plasma protein binding, and volume of distribution. Ticlodipine-associated TMA is discussed later in the chapter. For mushroom poisoning caused by *Amanita phalloides*, TPE is used, though some sources suggest that forced diuresis may be more effective.¹⁴⁴ There is a lack of clear evidence regarding the benefits of TPE in treating severe intoxications with tricyclic antidepressants, benzodiazepines, quinine, and phenytoin. While TPE is effective in removing drugs like L-thyroxine, verapamil, diltiazem, carbamazepine, theophylline, and heavy metals, the overall impact on total body toxin levels is often not clinically significant (Table e-65.3). The lack of controlled studies makes it challenging to establish definitive treatment recommendations for poisonings and overdoses. TPE has been used to treat severe reactions to wasp stings.¹⁴⁵ This study suggests that plasmapheresis may help reduce the severity of symptoms and improve outcomes in severe cases of wasp venom toxicity. Plasmapheresis is generally reserved for cases where other treatments are ineffective or there is presence of life-threatening toxicity. Before this, it is important to consider the availability of other treatment options (e.g., activated charcoal and hemodialysis) and comorbidities of patients.

LIVER DISEASE

Liver disease, especially in its advanced stages, can lead to life-threatening complications like hepatic encephalopathy, coagulopathy, and multiorgan failure. Plasmapheresis is employed in several liver-related conditions, particularly in scenarios where conventional treatments are ineffective or inadequate. In acute liver failure, there is rapid accumulation of albumin-bound toxins, as well as unbound

toxins, including aromatic amino acids, ammonia, endotoxin, indols, mercaptans, phenols, and other factors called damage-associated molecular patterns (DAMPs). These factors contribute to hepatic encephalopathy, reduced blood pressure, and reduced cerebral blood flow. TPE in combination with other modalities has shown reductions in some of these mediators leading to improved cerebral blood flow, mean arterial pressure, and other laboratory parameters. A randomized controlled trial involving 126 patients with acute liver failure showed that high-volume plasmapheresis compared with medical therapy alone improved transplant-free survival by similarly reducing systemic inflammation markers (84% vs. 26%) and removing toxic substances like cytokines and ammonia from the bloodstream (221.5 ± 96.9) versus (439 ± 385.6)).^{145a} Plasmapheresis was also associated with a higher 21-day transplant-free survival.

MARS is a liver support system used in liver failure, primarily to manage complications from hepatic encephalopathy, jaundice, and hepatorenal syndrome. MARS acts as a detoxification system, filtering out albumin-bound and water-soluble toxins that accumulate due to liver dysfunction. It combines dialysis with albumin to remove harmful substances like bilirubin and ammonia (see Fig. 65.3). Studies have shown that MARS can temporarily stabilize patients, providing a bridge to liver transplantation or spontaneous recovery. Hanish¹⁴⁶ reported 27 patients with severe ALF who received MARS therapy. Five patients with severe liver trauma had a 60% survival with recovery of liver and renal function. Thirteen patients received MARS as a bridge to transplant, of which 9 were transplanted with a 1-year survival of 78% (program overall survival 85% at 1 year). Nine patients with ALF from toxic ingestion received MARS as DT with liver recovery and survival in 67%. MARS therapy resulted in significant improvement in liver function, coagulation, incidence of encephalopathy, and creatinine. However, RELIEF randomized 189 patients with acute on chronic liver failure (ACLF) to MARS ($n = 95$) or standard therapy (SMT) ($n = 94$).^{146a} The 28-day survival was similar in the two groups in the intention-to-treat and per-protocol populations (60.7% vs. 58.9%; 60% vs. 59.2%, respectively). More recently, Gerth and colleagues compared 101 patients with ACLF who received standard medical treatment ($n = 54$) or standard medical treatment and the MARS ($n = 47$). A reduced 14-day mortality rate was observed in the MARS group (9.5% vs. 50.0% with standard medical treatment; $P = 0.004$), especially in patients with multiple organ failure.¹⁴⁷ In another study of patients awaiting liver transplant; the addition of 3 daily TPE procedures to standard of care in a randomized trial of 183 patients showed liver transplantation-free survival to hospital discharge in 59% of the TPE group versus 48% in the standard-care only group ($P < .001$).¹⁴⁸ However, there were no differences in the survival of the groups after liver transplantation. Some centers are using combined blood purification treatments (TPE + continuous renal replacement therapy [CRRT], TPE + hemofiltration, MARS [Molecular Adsorbent Recirculating System] + CRRT). This seems to have the rationale that the liver, as well as the kidney and other organ systems, are involved.

The Asian Pacific Association for the Study of the Liver (APASL) recommends the use of plasmapheresis in

combination with CRRT for patients with acute liver failure, as it offers better detoxification and hemodynamic support, particularly in patients with elevated Model for End-Stage Liver Disease scores. Although the data are conflicting, MARS is currently the only FDA-approved option in the United States for liver failure related to overdoses or poisonings and, if available, could provide critical support while awaiting liver transplantation or recovery from toxic injury.

WILSON DISEASE

Wilson disease (WD), a genetic disorder that leads to copper accumulation in the liver, brain, and other organs, can cause ALF. Plasmapheresis is used in WD as an adjunctive therapy, particularly in cases of severe hepatic and neurologic manifestations. Plasmapheresis reduces plasma copper levels, which is essential for patients with copper-induced acute liver failure. Hilal¹⁴⁹ demonstrated the use of plasmapheresis in treating WD, particularly during fulminant hepatic failure. Plasmapheresis provides a bridge before liver transplantation. The therapy improved mental status, renal function, and reduced bilirubin levels. This approach is vital for stabilizing patients during critical periods, when chelation therapy is insufficient or too slow.

PREGNANCY AND PLASMAPHERESIS

TPE in pregnancy is implemented on a case-by-case basis depending on pathophysiology. As with many indications, there are only case reports to inform decision making. Introduction of this procedure during pregnancy for diseases requiring that procedure has improved maternal and fetal survival rates. A summary of pregnancy-associated conditions where TPE may be considered is shown in [Table 65.4](#).

TPE can overall be safely performed during pregnancy. When comparing complications of TPE in pregnant versus nonpregnant counterparts, the side effect profile was similar in both groups except for maternal blood pressure drop. This also resulted in a drop in fetal heartbeat rate. However, when intravascular volume was maintained or repleted, side

effects were transient¹⁵⁰ and place the patient on her left side to avoid compression of the inferior vena cava by the gravid uterus. Other complications can occur due to allergy (additionally from replacement fluid of FFP), hypocalcemia, transitory cardiac arrhythmias, nausea, and blurred vision.

The first case of TPE used in pregnancy was in 1968¹⁵¹ in a pregnant woman with red cell alloimmunization. When maternal plasma contains an alloantibody against a red cell antigen carried by the fetus, hemolytic disease of the fetus and newborn (HDFN) occurs. The maternal IgG crosses the placenta and causes hemolysis of the fetal red cells. This leads to fetal anemia and, when severe enough, fetal death (hydrops fetalis). Usually, HDFN is secondary to anti-D (previously termed “Rh disease”), but it can be caused by a variety of red cell alloantibodies—such as anti-K, anti-C, anti-PP1Pk, and anti-E. TPE removes the maternal red cell alloantibody that causes HDFN. Therefore it is thought that TPE decreases the maternal antibody titer and, in turn, the amount transferred to the fetus, thereby protecting it from HDFN. Survival in severe cases of HDFN with the use of TPE and IVIg before intrauterine transfusion (before a gestation period of 20 weeks) is about 70%. However, with the routine prophylactic use of RhD immunoglobulin, HDFN rates have reduced considerably and thus the need for TPE.

TPE has an important role in the treatment of thrombotic-microangiopathy-related disorders in pregnancy. Until TPE was used for TTP during pregnancy, maternal survival was rare and the fetal mortality rate approached 80%. During pregnancy, 10% to 25% of the initial presenting episode of TTP occurs.¹⁵² In the past 2 decades, a series of reports have revealed the efficacy of first-line plasma exchange along with immunosuppression for TTP in pregnancy. Response rates are about 80% and there are minimal or no long-term sequelae. On the other hand, in TMA conditions such as atypical hemolytic uremic syndrome (aHUS), eculizumab, a complement inhibitor, should be used in addition to TPE for improved renal recovery outcomes.

TPE has also been safely carried out in pregnant patients myasthenic crisis, GBS, anti-GBM disease, SLE, multiple sclerosis, and acute fatty liver of pregnancy. In addition, LDL apheresis has been tried with varied success for conditions such as hypercholesterolemia/hypertriglyceridemia

Table 65.4 Indications for Plasma Exchange in Pregnancy-Related Conditions

Condition	Indication	Additional Notes
Thrombotic thrombocytopenic purpura (TTP)	Removes ADAMTS13 autoantibodies and replenishes ADAMTS13 enzyme with fresh frozen plasma (FFP)	A life-threatening emergency requiring prompt treatment. Typically combined with corticosteroids
Hemolysis, elevated liver enzymes, low platelets (HELLP)	Severe cases with multiorgan failure or atypical presentation	Limited evidence; TPE may be beneficial in refractory cases
Severe hypertriglyceridemia in pregnancy	When severe hypertriglyceridemia leads to acute pancreatitis	TPE removes triglycerides and prevents further organ damage
Autoimmune diseases (e.g., lupus, myasthenia gravis)	Treats severe flares, especially in life-threatening situations like lupus nephritis or myasthenic crisis.	TPE helps in antibody removal; adjunctive to immunosuppressive therapy
Refractory preeclampsia or eclampsia	Limited use in severe, refractory cases with multiorgan dysfunction	Insufficient evidence; only in exceptional situations and after delivery
Severe Rh isoimmunization	Reduces circulating maternal alloantibodies targeting fetal red cells	Typically reserved for severe cases where fetal transfusions are not sufficient

since alternative approaches may be teratogenic. TPE has also been used in acute hypertriglyceridemia-induced pancreatitis in pregnancy and as a prophylactic measure for the prevention of pancreatitis in pregnant women with known hypertriglyceridemia.

THERAPEUTIC PLASMA EXCHANGE IN PEDIATRICS

Data on the efficacy of TPE are even more sparse in other forms of vasculitis and in children. TPE is relatively safe in children, with pruritis and urticaria the most common complications, but more serious complications, including infection, have been reported.²⁴ In patients with Churg-Strauss syndrome or polyarteritis nodosa, TPE does not appear to be of benefit in addition to cyclophosphamide and steroids.¹⁵³ In children with ANCA-associated vasculitis, the removal of circulating ANCA through TPE is a short-term measure reserved for severe or refractory disease.¹⁵⁴ There is little evidence for or against TPE in other causes of RPGN, although there has been one report of benefit in children with rapidly progressive glomerulonephritis from Henoch-Schönlein purpura.¹⁵⁵ One study has suggested that TPE may be beneficial in kidney transplant recipients with recurrent Henoch-Schönlein purpura nephritis, but there have been no prospective studies or protocols indicating the optimal therapeutic regimen in this group of patients.¹⁵⁶

CONCLUSION

The use of TPE to treat a variety of conditions has grown significantly in recent years. In some cases, the rationale and benefit are supported by clinical studies, but in many cases, the benefits have not been well established. Nevertheless, the concept of removing plasma containing pathogenic antibodies is now well established. Additional studies are needed to determine the potential benefits for TPE in these other conditions.

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KEY REFERENCES

- Connelly-Smith L, Alquist CR, Aqui NA, et al. Guidelines on the use of therapeutic apheresis in clinical practice - evidence-based approach from the Writing Committee of the American Society for apheresis: the ninth special issue. *J Clin Apher*. 2023;38(2):77–278.
- Mokrzycki MH, Kaplan AA. Therapeutic plasma exchange: complications and management. *Am J Kidney Dis*. 1994;23(6):817–827.
- McAdoo SP, Pusey CD. Anti-glomerular basement membrane disease. *Clin J Am Soc Nephrol*. 2017;12(7):1162–1172.
- Lee JB, Trew AJ, Ross AJ, Pusey CD. Long-term outcome of anti-glomerular basement membrane antibody disease treated with plasma exchange and immunosuppression. *Ann Intern Med*. 2001;134(11):955–961.
- Huart A, Josse AG, Chauveau D, et al. Outcomes of patients with Goodpasture syndrome: a nationwide cohort-based study from the French Society of Hemapheresis. *J Autoimmun*. 2016;73:24–29.
- Zahir Z, Wani AS, Prasad N, Jain M. Clinicopathological characteristics and predictors of poor outcome in anti-glomerular basement membrane disease - a fifteen year single center experience. *Ren Fail*. 2021;43(1):79–89.
- Kuang H, Jiang N, Jia XY, Cui Z, Zhao MH. Epidemiology, clinical features, risk factors, and outcomes in anti-glomerular basement membrane disease: a systematic review and meta-analysis. *Autoimmun Rev*. 2024;23(4):103531.
- Biesenbach P, Kain R, Derfler K, et al. Long-term outcome of anti-glomerular basement membrane antibody disease treated with immunoadsorption. *PLoS One*. 2014;9(7):e103568.
- Sallee M, Resseguier N, Crepin T, et al. Immunoadsorption and plasma exchange are comparable in anti-neutrophil cytoplasmic antibodies or anti-glomerular basement membrane removal kinetics. *Kidney Int Rep*. 2024;9(9):2767–2773.
- Chen BY, Chan TM, Leung KN. Recurrent glomerulonephritis after kidney transplantation. *Am J Transplant*. 2006;6(11):2557–2564.
- Kidney disease: improving global outcomes glomerular diseases work G. KDIGO 2021 clinical practice guideline for the management of glomerular diseases. *Kidney Int*. 2021;100(4S):S1–S276.
- Son SC, Bae S, Lee SV, et al. Recurrence and outcome of anti-glomerular basement membrane glomerulonephritis after kidney transplantation. *Kidney Int Rep*. 2021;6(7):1890–1898.
- Lockwood CM, Pinching AJ, Sweny P, et al. Plasma-exchange and immunosuppression in the treatment of fulminating immune-complex crescentic nephritis. *Lancet*. 1977;1(8002):63–67.
- Pusey CD, Rees AJ, Evans DJ, Peters DK, Lockwood CM. Plasma exchange in focal necrotizing glomerulonephritis without anti-GBM antibodies. *Kidney Int*. 1991;40(4):757–763.
- Cole E, Cattran D, Magil A, et al. A prospective randomized trial of plasma exchange as additive therapy in idiopathic crescentic glomerulonephritis. The Canadian Apheresis Study Group. *Am J Kidney Dis*. 1992;20(3):261–269.
- Walters G, Willis NS, Craig JC. Interventions for renal vasculitis in adults. *Cochrane Database Syst Rev*. 2015;(9):CD003232.
- Walsh M, Merkel PA, Peh CA, et al. Plasma exchange and glucocorticoids in severe ANCA-associated vasculitis. *N Engl J Med*. 2020;382(7):622–631.
- Klemmer PJ, Chalermkulrat W, Reif MS, Hogan SL, Henke DC, Falk RJ. Plasmapheresis therapy for diffuse alveolar hemorrhage in patients with small-vessel vasculitis. *Am J Kidney Dis*. 2003;42(6):1149–1153.
- Rovin BH, Furie R, Latinis K, et al. Efficacy and safety of rituximab in patients with active proliferative lupus nephritis: the Lupus Nephritis Assessment with Rituximab study. *Arthritis Rheum*. 2012;64(4):1215–1226.
- Lewis EJ, Hunsicker LG, Lan SP, Rohde RD, Lachin JM. A controlled trial of plasmapheresis therapy in severe lupus nephritis. The Lupus Nephritis Collaborative Study Group. *N Engl J Med*. 1992;326(21):1373–1379.
- Wallace DJ, Goldfinger D, Pepkowitz SH, et al. Randomized controlled trial of pulse/synchronization cyclophosphamide/apheresis for proliferative lupus nephritis. *J Clin Apher*. 1998;13(4):163–166.
- Hutchison CA, Cockwell P, Moroz V, et al. High cutoff versus high-flux haemodialysis for myeloma cast nephropathy in patients receiving bortezomib-based chemotherapy (EuLITE): a phase 2 randomised controlled trial. *Lancet Haematol*. 2019;6(4):e217–e228.
- Premuzic V, Batinic J, Roncovic P, Basic-Jukic N, Nemet D, Jelakov B. Role of plasmapheresis in the management of acute kidney injury in patients with multiple myeloma: should we abandon it? *Ther Apher Dial*. 2018;22(1):79–86.
- Clark WF, Stewart AK, Rock GA, et al. Plasma exchange when myeloma presents as acute renal failure: a randomized, controlled trial. *Ann Intern Med*. 2007;146(6):471. published correction appears in.
- Hutchison CA, Harding S, Mead G, et al. Serum free-light chain removal by high cutoff hemodialysis: optimizing removal and supportive care. *Artif Organs*. 2008;32(12):910–917.
- Sens F, Chaintreuil D, Jolivet A, et al. Effectiveness of IHD with adsorptive PMMA membrane in myeloma cast nephropathy: a cohort study. *Am J Nephrol*. 2017;46(5):355–363.
- Gertz MA. Waldenström macroglobulinemia: 2023 update on diagnosis, risk stratification, and management. *Am J Hematol*. 2023;98(2):348–358.
- George JN, Nester CM. Syndromes of thrombotic microangiopathy. *N Engl J Med*. 2014;371(19):1847–1848.

82. Joly BS, Coppo P, Veyradier A. Thrombotic thrombocytopenic purpura. *Blood*. 2017;129(21):2836–2846.
85. Bell WR, Braine HG, Ness PM, Kickler TS. Improved survival in thrombotic thrombocytopenic purpura-hemolytic uremic syndrome. Clinical experience in 108 patients. *N Engl J Med*. 1991;325(6):398–403.
88. Rock GA, Shumak KH, Buskard NA, et al. Comparison of plasma exchange with plasma infusion in the treatment of thrombotic thrombocytopenic purpura. Canadian Apheresis Study Group. *N Engl J Med*. 1991;325(6):393–397.
91. Schwartz J, Padmanabhan A, Aquilino N, et al. Guidelines on the use of therapeutic apheresis in clinical practice-evidence-based approach from the Writing Committee of the American Society for apheresis: the seventh special issue. *J Clin Apher*. 2016;31(3):149–162.
94. Kielstein JT, Beutel G, Fleig S, et al. Best supportive care and therapeutic plasma exchange with or without eculizumab in Shiga-toxin-producing *E. coli* O104:H4 induced haemolytic-uraemic syndrome: an analysis of the German STEC-HUS registry. *Nephrol Dial Transplant*. 2012;27(10):3807–3815.
97. D'Agati VD, Kaskel FJ, Falk RJ. Focal segmental glomerulosclerosis. *N Engl J Med*. 2011;365(25):2398–2411.
100. Hayek SS, Sever S, Ko YA, et al. Soluble urokinase receptor and chronic kidney disease. *N Engl J Med*. 2015;373(20):1916–1925.
102. Gohh RY, Yango AF, Morrissey PE, et al. Preemptive plasmapheresis and recurrence of FSGS in high-risk renal transplant recipients. *Am J Transplant*. 2005;5(12):2907–2912.
105. Ohta T, Kawaguchi H, Hattori M, et al. Effect of pre-and post-operative plasmapheresis on posttransplant recurrence of focal segmental glomerulosclerosis in children. *Transplantation*. 2001;71(5):628–633.
107. Veissi ST, Smeets B, van Wijk JAE, van der Velden T, van den Heuvel LPWJ, Schreuder MF. Plasma exchange or immunoadsorption for recurrent focal segmental glomerulosclerosis: clear differences in vitro. *Nephrol Dial Transplant*. 2022;37(11):2293–2294.
113. Ohkubo A, Okado T. Selective plasma exchange. *Transfus Apher Sci*. 2017;56(5):657–660. <https://doi.org/10.1016/j.transci.2017.08.010>.
114. Hanaoka A, Naganuma T, Kabata D, et al. Selective plasma exchange in ABO-incompatible kidney transplantation: comparison of substitution with albumin and partial substitution with fresh frozen plasma. *Sci Rep*. 2020;10(1):1434.
115. Stegall MD, Gloor J, Winters JL, Moore SB, Degoe S. A comparison of plasmapheresis versus high-dose IVIG desensitization in renal allograft recipients with high levels of donor specific alloantibody. *Am J Transplant*. 2006;6(2):346–351.
117. Montgomery RA, Lonze BE, King KE, et al. Desensitization in HLA-incompatible kidney recipients and survival. *N Engl J Med*. 2011;365(4):318–326.
119. Pascual M, Saidman S, Tolckoff-Rubin N, et al. Plasma exchange and tacrolimus-mycophenolate rescue for acute humoral rejection in kidney transplantation [published correction appears in *Transplantation* 1999 Feb 15;67(3):495]. *Transplantation*. 1998;66(11):1460–1464.
120. Lefaucheur C, Nochy D, Andrade J, et al. Comparison of combination Plasmapheresis/IVIg/anti-CD20 versus high-dose IVIg in the treatment of antibody-mediated rejection. *Am J Transplant*. 2009;9(5):1099–1107.
133. van der Meche FG, Schmitz PI. A randomized trial comparing intravenous immune globulin and plasma exchange in Guillain-Barre syndrome. Dutch Guillain-Barre Study Group. *N Engl J Med*. 1992;326(17):1123–1129.
146. Hanish SI, Stein DM, Scalea JR, et al. Molecular adsorbent recirculating system effectively replaces hepatic function in severe acute liver failure. *Ann Surg*. 2017;266(4):677–684.
152. Scully M, Starke R, Lee R, Mackie I, Machin S, Cohen H. Successful management of pregnancy in women with a history of thrombotic thrombocytopenic purpura. *Blood Coagul Fibrinolysis*. 2006;17(6):459–463.

REFERENCES

- Connelly-Smith L, Alquist CR, Aqui NA, et al. Guidelines on the use of therapeutic apheresis in clinical practice - evidence-based approach from the Writing Committee of the American Society for apheresis: the ninth special issue. *J Clin Apher.* 2023;38(2):77–278.
- Kambic HE, Nose Y. Historical perspective on plasmapheresis. *Ther Apher.* 1997;1(1):83–108.
- Abel JJ, Rowntree LG, Turner BB. On the removal of diffusible substances from the circulating blood by means of dialysis. Transactions of the Association of American Physicians, 1913. *Transfus Sci.* 1990;11(2):164–165.
- Schwab PJ, Fahey JL. Treatment of Waldenstrom's macroglobulinemia by plasmapheresis. *N Engl J Med.* 1960;263:574–579.
- Reeves HM, Winters JL. The mechanisms of action of plasma exchange. *Br J Haematol.* 2014;164(3):342–351.
- Zarkovic M, Kwaan HC. Correction of hyperviscosity by apheresis. *Semin Thromb Hemost.* 2003;29(5):535–542.
- Hafer C, Golla P, Gericke M, et al. Membrane versus centrifuge-based therapeutic plasma exchange: a randomized prospective crossover study. *Int Urol Nephrol.* 2016;48(1):133–138.
- Stegmayr B, Wikdahl AM. Access in therapeutic apheresis. *Ther Apher Dial.* 2003;7(2):209–214.
- Madore F, Lazarus JM, Brady HR. Therapeutic plasma exchange in renal diseases. *J Am Soc Nephrol.* 1996;7(3):367–386.
- Reimann PM, Mason PD. Plasmapheresis: technique and complications. *Intensive Care Med.* 1990;16(1):3–10.
- Ziai F, Benesch T, Kodras K, Neumann I, Dimopoulos-Xicki L, Haas M. The effect of oral anticoagulation on clotting during hemodialysis. *Kidney Int.* 2005;68(2):862–866.
- Linenberger ML, Price TH. Use of cellular and plasma apheresis in the critically ill patient: part 1: technical and physiological considerations. *J Intensive Care Med.* 2005;20(1):18–27.
- Hassell K. The management of patients with heparin-induced thrombocytopenia who require anticoagulant therapy. *Chest.* 2005;127(2 Suppl):1s–8s.
- Lowentritt JE, Kahn MJ, Batuman V. Use of recombinant hirudin as an anticoagulant in plasmapheresis in a patient with heparin-induced thrombocytopenia. *Nephrol Dial Transplant.* 2002;17(8):1533–1534.
- Le Conte P, Nicolas F, Adjou C, N'Guyen JM, Billaud E, Moreau P. Replacement fluids in plasmapheresis: cross-over comparative study. *Intensive Care Med.* 1997;23(3):342–344.
- Brecher ME, Owen HG, Bandarenko N. Alternatives to albumin: starch replacement for plasma exchange. *J Clin Apher.* 1997;12(3):146–153.
- Agreda-Vasquez GP, Espinosa-Poblano I, Sanchez-Guerrero SA, et al. Starch and albumin mixture as replacement fluid in therapeutic plasma exchange is safe and effective. *J Clin Apher.* 2008;23(5):163–167.
- Arslan O, Tek I, Arat M, Ciftci I, Ilhan O. Effects of replacement fluids on plasma viscosity used for therapeutic plasma exchange. *Ther Apher Dial.* 2004;8(2):144–147.
- Auwerda JJ, Wilson JH, Sonneveld P. Foamy macrophage syndrome due to hydroxyethyl starch replacement: a severe side effect in plasmapheresis. *Ann Intern Med.* 2002;137(12):1013–1014.
- Auwerda JJ, Leebeek FW, Wilson JH, van Diggelen OP, Lam KH, Sonneveld P. Acquired lysosomal storage caused by frequent plasmapheresis procedures with hydroxyethyl starch. *Transfusion.* 2006;46(10):1705–1711.
- Mokrzycki MH, Kaplan AA. Therapeutic plasma exchange: complications and management. *Am J Kidney Dis.* 1994;23(6):817–827.
- Lu Q, Nedelcu E, Ziman A, Bumerts P, Fernando L, Schiller G. Standardized protocol to identify high-risk patients undergoing therapeutic apheresis procedures. *J Clin Apher.* 2008;23(3):111–115.
- Kiprov DD, Golden P, Rohe R, Smith S, Hofmann J, Hunnicutt J. Adverse reactions associated with mobile therapeutic apheresis: analysis of 17,940 procedures. *J Clin Apher.* 2001;16(3):130–133.
- Lu J, Zhang L, Xia C, Tao Y. Complications of therapeutic plasma exchange: a retrospective study of 1201 procedures in 435 children. *Medicine.* 2019;98(50):e18308.
- Kankirawatana S, Huang ST, Marques MB. Continuous infusion of calcium gluconate in 5% albumin is safe and prevents most hypocalcemic reactions during therapeutic plasma exchange. *J Clin Apher.* 2007;22(5):265–269.
- Marques MB, Huang ST. Patients with thrombotic thrombocytopenic purpura commonly develop metabolic alkalosis during therapeutic plasma exchange. *J Clin Apher.* 2001;16(3):120–124.
- Keller AJ, Chirnside A, Urbaniak SJ. Coagulation abnormalities produced by plasma exchange on the cell separator with special reference to fibrinogen and platelet levels. *Br J Haematol.* 1979;42(4):593–603.
- Yeh JH, Chiu HC. Coagulation abnormalities in serial double-filtration plasmapheresis. *J Clin Apher.* 2001;16(3):139–142.
- Brecher ME. Plasma exchange: why we do what we do. *J Clin Apher.* 2002;17(4):207–211.
- Pohl MA, Lan SP, Berl T. Plasmapheresis does not increase the risk for infection in immunosuppressed patients with severe lupus nephritis. The Lupus Nephritis Collaborative Study Group. *Ann Intern Med.* 1991;114(11):924–929.
- Owen HG, Brecher ME. Atypical reactions associated with use of angiotensin-converting enzyme inhibitors and apheresis. *Transfusion.* 1994;34(10):891–894.
- Weinstein R. Hypocalcemic toxicity and atypical reactions in therapeutic plasma exchange. *J Clin Apher.* 2001;16(4):210–211.
- Tanabe K. Double-filtration plasmapheresis. *Transplantation.* 2007;84(12 Suppl):S30–S32.
- Stegmayr BG. A survey of blood purification techniques. *Transfus Apher Sci.* 2005;32(2):209–220.
- Savage CO, Pusey CD, Bowman C, Rees AJ, Lockwood CM. Antiglomerular basement membrane antibody mediated disease in the British Isles 1980–4. *Br Med J.* 1986;292(6516).
- Gu QH, Xie LJ, Jia XY, et al. Fever and prodromal infections in anti-glomerular basement membrane disease. *Nephrology.* 2018;23(5):476–482.
- Donaghy M, Rees AJ. Cigarette smoking and lung haemorrhage in glomerulonephritis caused by autoantibodies to glomerular basement membrane. *Lancet.* 1983;2(8364):1390–1393.
- McAdoo SP, Pusey CD. Anti-glomerular basement membrane disease. *Clin J Am Soc Nephrol.* 2017;12(7):1162–1172.
- Lockwood CM, Rees AJ, Pearson TA, Evans DJ, Peters DK, Wilson CB. Immunosuppression and plasma-exchange in the treatment of Goodpasture's syndrome. *Lancet.* 1976;1(7962):711–715.
- Levy JB, Turner AN, Rees AJ, Pusey CD. Long-term outcome of anti-glomerular basement membrane antibody disease treated with plasma exchange and immunosuppression. *Ann Intern Med.* 2001;134(11):1033–1042.
- Cui Z, Zhao J, Jia XY, et al. Anti-glomerular basement membrane disease: outcomes of different therapeutic regimens in a large single-center Chinese cohort study. *Medicine.* 2011;90(5):303–311.
- Huart A, Josse AG, Chauveau D, et al. Outcomes of patients with Goodpasture syndrome: a nationwide cohort-based study from the French Society of Hemapheresis. *J Autoimmun.* 2016;73:24–29.
- Zahir Z, Wani AS, Prasad N, Jain M. Clinicopathological characteristics and predictors of poor outcome in anti-glomerular basement membrane disease - a fifteen year single center experience. *Ren Fail.* 2021;43(1):79–89.
- Kuang H, Jiang N, Jia XY, Cui Z, Zhao MH. Epidemiology, clinical features, risk factors, and outcomes in anti-glomerular basement membrane disease: a systematic review and meta-analysis. *Autoimmun Rev.* 2024;23(4):103531.
- Johnson JP, Moore Jr J, Austin 3rd HA, Balow JE, Antonovych TT, Wilson CB. Therapy of anti-glomerular basement membrane antibody disease: analysis of prognostic significance of clinical, pathologic and treatment factors. *Medicine.* 1985;64(4):219–227.
- Biesenbach P, Kain R, Derfler K, et al. Long-term outcome of anti-glomerular basement membrane antibody disease treated with immunoadsorption. *PLoS One.* 2014;9(7):e103568.
- Sallee M, Resseguier N, Crepin T, et al. Immunoadsorption and plasma exchange are comparable in anti-neutrophil cytoplasmic antibodies or anti-glomerular basement membrane removal kinetics. *Kidney Int Rep.* 2024;9(9):2767–2773.
- Choy BY, Chan TM, Lai KN. Recurrent glomerulonephritis after kidney transplantation. *Am J Transplant.* 2006;6(11):2535–2542.
- Kidney Disease. Improving global outcomes glomerular diseases work G. KDIGO 2021 clinical practice guideline for the management of glomerular diseases. *Kidney Int.* 2021;100(4S):S1–S276.

50. Coche S, Sprangers B, Van Laecke S, et al. Recurrence and outcome of anti-glomerular basement membrane glomerulonephritis after kidney transplantation. *Kidney Int Rep.* 2021;6(7).
51. Lockwood CM, Pinching AJ, Sweny P, et al. Plasma-exchange and immunosuppression in the treatment of fulminating immune-complex crescentic nephritis. *Lancet.* 1977;1(8002):63–67.
52. Pusey CD, Rees AJ, Evans DJ, Peters DK, Lockwood CM. Plasma exchange in focal necrotizing glomerulonephritis without anti-GBM antibodies. *Kidney Int.* 1991;40(4):757–763.
53. Cole E, Cattran D, Magil A, et al. A prospective randomized trial of plasma exchange as additive therapy in idiopathic crescentic glomerulonephritis. The Canadian Apheresis Study Group. *Am J Kidney Dis.* 1992;20(3):261–269.
54. Jayne DR, Gaskin G, Rasmussen N, et al. Randomized trial of plasma exchange or high-dosage methylprednisolone as adjunctive therapy for severe renal vasculitis. *J Am Soc Nephrol.* 2007;18(7):2180–2188.
55. Walters G, Willis NS, Craig JC. Interventions for renal vasculitis in adults. *Cochrane Database Syst Rev.* 2015;(9):CD003232.
56. Walsh M, Merkel PA, Peh CA, et al. Plasma exchange and glucocorticoids in severe ANCA-associated vasculitis. *N Engl J Med.* 2020;382(7):622–631.
57. Yates M, Watts RA, Bajema IM, et al. EULAR/ERA-EDTA recommendations for the management of ANCA-associated vasculitis. *Ann Rheum Dis.* 2016;75(9):1583–1594.
58. Gallagher H, Kwan JT, Jayne DR. Pulmonary renal syndrome: a 4-year, single-center experience. *Am J Kidney Dis.* 2002;39(1):42–47.
59. Klemmer PJ, Chalermkulrat W, Reif MS, Hogan SL, Henke DC, Falk RJ. Plasmapheresis therapy for diffuse alveolar hemorrhage in patients with small-vessel vasculitis. *Am J Kidney Dis.* 2003;42(6):1149–1153.
60. Szpiro WM. Plasma exchange in antineutrophil cytoplasmic antibody-associated vasculitis—a 25-year perspective. *Nephrol Dial Transplant.* 2015;30(suppl 1):i146–i149.
61. Rovin BH, Furie R, Latinis K, et al. Efficacy and safety of rituximab in patients with active proliferative lupus nephritis: the Lupus Nephritis Assessment with Rituximab study. *Arthritis Rheum.* 2012;64(4):1215–1226.
62. Jones JV, Cumming RH, Bucknall RC, Asplin CM. Plasmapheresis in the management of acute systemic lupus erythematosus? *Lancet.* 1976;1(7962):709–711.
63. Lewis EJ, Hunsicker LG, Lan SP, Rohde RD, Lachin JM. A controlled trial of plasmapheresis therapy in severe lupus nephritis. The Lupus Nephritis Collaborative Study Group. *N Engl J Med.* 1992;326(21):1373–1379.
64. Wallace DJ, Goldfinger D, Pepkowitz SH, et al. Randomized controlled trial of pulse/synchronization cyclophosphamide/apheresis for proliferative lupus nephritis. *J Clin Apher.* 1998;13(4):163–166.
65. Danieli MG, Palmieri C, Salvi A, Refe MC, Strusi AS, Danieli G. Synchronised therapy and high-dose cyclophosphamide in proliferative lupus nephritis. *J Clin Apher.* 2002;17(2):72–77.
66. Gulli F, Santini SA, Napodano C, et al. Cryoglobulin test and cryoglobulinemia hepatitis C-virus related. *Mediterr J Hematol Infect Dis.* 2017;9(1).
67. Naciri BH, Banza AT, Terrec F, et al. Cryoglobulinemia and double-filtration plasmapheresis: personal experience and literature review. *Ther Apher Dial.* 2023;27(1):159–169.
68. Marson P, Monti G, Montani F, et al. Apheresis treatment of cryoglobulinemic vasculitis: a multicentre cohort study of 159 patients. *Transfus Apher Sci.* 2018;57(5):639–645.
- 68a. Chute DF, Chung RT, Sise ME. Direct-acting antiviral therapy for hepatitis C virus infection in the kidney transplant recipient. *Kidney Int.* 2018;93(3):560–567.
69. Leung N, Nasr SH. Myeloma-related kidney disease. *Adv Chronic Kidney Dis.* 2014;21(1):36–47.
70. Hutchison CA, Cockwell P, Moroz V, et al. High cutoff versus high-flux haemodialysis for myeloma cast nephropathy in patients receiving bortezomib-based chemotherapy (EuLITE): a phase 2 randomised controlled trial. *Lancet Haematol.* 2019;6(4):e217–e228.
71. Premuzic V, Batinic J, Roncovic P, Basic-Jukic N, Nemet D, Jelakovac B. Role of plasmapheresis in the management of acute kidney injury in patients with multiple myeloma: should we abandon it? *Ther Apher Dial.* 2018;22(1):79–86.
72. Clark WF, Stewart AK, Rock GA, et al. Plasma exchange when myeloma presents as acute renal failure: a randomized, controlled trial [published correction appears in. *Ann Intern Med.* 2007;146(6):471.
- 72a. Hutchison CA, Cockwell P, Reid S, et al. Efficient removal of immunoglobulin free light chains by hemodialysis for multiple myeloma: In vitro and in vivo studies. *J Am Soc Nephrol.* 2007;18(3):886–895.
73. Hutchison CA, Harding S, Mead G, et al. Serum free-light chain removal by high cutoff hemodialysis: optimizing removal and supportive care. *Artif Organs.* 2008;32(12):910–917.
74. Berni Wennekers A, Martín Azara MP, Dourdil SV, et al. Thirteen treated of acute renal failure secondary to multiple myeloma with high cut off filters. Trece tratamientos de la insuficiencia renal aguda secundaria a mieloma múltiple con filtros de high cut off. *Nefrologia.* 2016;36(4):418–426.
75. Huang J, Qiu F, Zhang H, Shen X, Lin X. Clinical effects of continuous veno-venous hemofiltration combined with hemoperfusion for the treatment of multiple myeloma complicated with acute kidney injury. *Pak J Med Sci.* 2023;39(2):344–348.
76. Sens F, Chaintreuil D, Jolivet A, et al. Effectiveness of IHD with adsorptive PMMA membrane in myeloma cast nephropathy: a cohort study. *Am J Nephrol.* 2017;46(5):355–363.
77. Abeykoon JP, Zanwar S, Ansell SM, et al. Predictors of symptomatic hyperviscosity in Waldenström macroglobulinemia. *Am J Hematol.* 2018;93(11):1384–1393.
78. Gertz MA. Waldenström macroglobulinemia: 2023 update on diagnosis, risk stratification, and management. *Am J Hematol.* 2023;98(2):348–358.
79. George JN, Nester CM. Syndromes of thrombotic microangiopathy. *N Engl J Med.* 2014;371(19):1847–1848.
80. Appel GB. Thrombotic microangiopathies: similar presentations, different therapies. *Cleve Clin J Med.* 2017;84(2):114–130.
81. Murrin RJ, Murray JA. Thrombotic thrombocytopenic purpura: aetiology, pathophysiology and treatment. *Blood Rev.* 2006;20(1):51–60.
82. Joly BS, Coppo P, Veyradier A. Thrombotic thrombocytopenic purpura. *Blood.* 2017;129(21):2836–2846.
83. Mariotte E, Azoulay E, Galicier L, et al. Epidemiology and pathophysiology of adulthood-onset thrombotic microangiopathy with severe ADAMTS13 deficiency (thrombotic thrombocytopenic purpura): a cross-sectional analysis of the French national registry for thrombotic microangiopathy. *Lancet Haematol.* 2016;3(5):e237–e245.
84. Byrnes JJ, Khurana M. Treatment of thrombotic thrombocytopenic purpura with plasma. *N Engl J Med.* 1977;297(25):1386–1389.
85. Bell WR, Braine HG, Ness PM, Kickler TS. Improved survival in thrombotic thrombocytopenic purpura-hemolytic uremic syndrome. Clinical experience in 108 patients. *N Engl J Med.* 1991;325(6):398–403.
86. Jamme M, Rondeau E. The PLASMIC score for thrombotic thrombocytopenic purpura. *Lancet Haematol.* 2017;4(4):e148–e149.
87. Payday K, Banwell E, Tong J, Chen Y, Cuker A. Diagnostic accuracy of the PLASMIC score in patients with suspected thrombotic thrombocytopenic purpura: a systematic review and meta-analysis. *Transfusion.* 2020;60(9):2047–2057.
88. Rock GA, Shumak KH, Buskard NA, et al. Comparison of plasma exchange with plasma infusion in the treatment of thrombotic thrombocytopenic purpura. Canadian Apheresis Study Group. *N Engl J Med.* 1991;325(6):393–397.
89. Scully M, Hunt BJ, Benjamin S, et al. Guidelines on the diagnosis and management of thrombotic thrombocytopenic purpura and other thrombotic microangiopathies. *Br J Haematol.* 2012;158(3):323–335.
90. Mehmood T, Taylor M, Winters JL. Management of thrombotic microangiopathic hemolytic anemias with therapeutic plasma exchange: when it works and when it does not. *Hematol Oncol Clin North Am.* 2016;30(3):679–694.
91. Schwartz J, Padmanabhan A, Aqui N, et al. Guidelines on the use of therapeutic apheresis in clinical practice-evidence-based approach from the Writing Committee of the American Society for apheresis: the seventh special issue. *J Clin Apher.* 2016;31(3):149–162.

92. Al-Nouri ZL, Reese JA, Terrell DR, Vesely SK, George JN. Drug-induced thrombotic microangiopathy: a systematic review of published reports. *Blood, J Am Soc Hematol*. 2015;125(4):616–618.
93. Bennett CL, Weinberg PD, Rozenberg-Ben-Dror K, Yarnold PR, Kwaan HC, Green D. Thrombotic thrombocytopenic purpura associated with ticlopidine. A review of 60 cases. *Ann Intern Med*. 1998;128(7):541–544.
94. Kielstein JT, Beutel G, Fleig S, et al. Best supportive care and therapeutic plasma exchange with or without eculizumab in Shiga-toxin-producing *E. coli* O104:H4 induced haemolytic-uraemic syndrome: an analysis of the German STEC-HUS registry. *Nephrol Dial Transplant*. 2012;27(10):3807–3815.
95. Kitiyakara C, Eggers P, Kopp JB. Twenty-one-year trend in ESRD due to focal segmental glomerulosclerosis in the United States. *Am J Kidney Dis*. 2004;44(5):815–825.
96. Uffing A, Pérez-Sáez MJ, Mazzali M, et al. Recurrence of FSGS after kidney transplantation in adults. *Clin J Am Soc Nephrol*. 2020;15(2):247–256.
97. D'Agati VD, Kaskel FJ, Falk RJ. Focal segmental glomerulosclerosis. *N Engl J Med*. 2011;365(25):2398–2411.
98. Roja JR, Pérez M, Hurtado A, Asato C. Factores pronósticos de supervivencia renal en glomerulosclerosis focal y segmentaria primaria [Factors predicting for renal survival in primary focal segmental glomerulosclerosis]. *Nefrologia*. 2008;28(4):439–446.
99. Wei C, El Hindi S, Li J, et al. Circulating urokinase receptor as a cause of focal segmental glomerulosclerosis. *Nat Med*. 2011;17(8):952–960.
100. Hayek SS, Sever S, Ko YA, et al. Soluble urokinase receptor and chronic kidney disease. *N Engl J Med*. 2015;373(20):1916–1925.
101. Zimmerman SW. Plasmapheresis and dipyridamole for recurrent focal glomerular sclerosis. *Nephron*. 1985;40(2):241–245.
102. Gohh RY, Yango AF, Morrissey PE, et al. Preemptive plasmapheresis and recurrence of FSGS in high-risk renal transplant recipients. *Am J Transplant*. 2005;5(12):2907–2912.
103. Reynolds BC, Lamb A, Jones CA, Yadav P, Tyerman KS, Geddes CC. UK experience of ofatumumab in recurrence of focal segmental glomerulosclerosis post-kidney transplant. *Pediatr Nephrol*. 2022;37(1):199–207.
104. Hansrivijit P, Puthenpura MM, Ghahramani N. Efficacy of abatacept treatment for focal segmental glomerulosclerosis and minimal change disease: a systematic review of case reports, case series, and observational studies. *Clin Nephrol*. 2020;94(3):117–126.
105. Ohta T, Kawaguchi H, Hattori M, et al. Effect of pre- and post-operative plasmapheresis on posttransplant recurrence of focal segmental glomerulosclerosis in children. *Transplantation*. 2001;71(5):628–633.
106. Ponticelli C. Recurrence of focal segmental glomerular sclerosis (FSGS) after renal transplantation. *Nephrol Dial Transplant*. 2010;25(1):25–31.
107. Veissi ST, Smeets B, van Wijk JAE, van der Velden T, van den Heuvel LPWJ, Schreuder MF. Plasma exchange or immunoadsorption for recurrent focal segmental glomerulosclerosis: clear differences in vitro. *Nephrol Dial Transplant*. 2022;37(11):2293–2294.
108. Alexandre GP, De Bruyere M, Squifflet JP, Moriau M, Latinne D, Pirson Y. Human ABO-incompatible living donor renal homo-grafts. *Neth J Med*. 1985;28(6):231–234.
109. Takahashi K, Saito K. ABO-incompatible kidney transplantation. *Transplant Rev*. 2013;27(1):1–8.
110. Montgomery RA, Locke JE, King KE, et al. ABO incompatible renal transplantation: a paradigm ready for broad implementation. *Transplantation*. 2009;87(8):1246–1255.
111. Montgomery JR, Berger JC, Warren DS, James NT, Montgomery RA, Segev DL. Outcomes of ABO-incompatible kidney transplantation in the United States. *Transplantation*. 2012;93(6):603–609.
112. Barnett AN, Manook M, Nagendran M, et al. Tailored desensitization strategies in ABO blood group antibody incompatible renal transplantation. *Transpl Int*. 2014;27(2):187–196.
113. Ohkubo A, Okado T. Selective plasma exchange. *Transfus Apher Sci*. 2017;56(5):657–660.
114. Hanaoka A, Naganuma T, Kabata D, et al. Selective plasma exchange in ABO-incompatible kidney transplantation: comparison of substitution with albumin and partial substitution with fresh frozen plasma. *Sci Rep*. 2020;10(1):1434.
115. Stegall MD, Gloor J, Winters JL, Moore SB, DeGoe S. A comparison of plasmapheresis versus high-dose IVIG desensitization in renal allograft recipients with high levels of donor specific alloantibody. *Am J Transplant*. 2006;6(2):346–351.
116. Thielke JJ, West-Thielke PM, Herren HL, et al. Living donor kidney transplantation across positive crossmatch: the University of Illinois at Chicago experience. *Transplantation*. 2009;87(2):268–273.
117. Montgomery RA, Lonze BE, King KE, et al. Desensitization in HLA-incompatible kidney recipients and survival. *N Engl J Med*. 2011;365(4):318–326.
118. Magee CC. Transplantation across previously incompatible immunological barriers. *Transpl Int*. 2006;19(2):87–97.
119. Pascual M, Saidman S, Tolkoff-Rubin N, et al. Plasma exchange and tacrolimus-mycophenolate rescue for acute humoral rejection in kidney transplantation [published correction appears in *Transplantation* 1999 Feb 15;67(3):495]. *Transplantation*. 1998;66(11):1460–1464.
120. Lefaucheur C, Nochy D, Andrade J, et al. Comparison of combination Plasmapheresis/IVIg/anti-CD20 versus high-dose IVIg in the treatment of antibody-mediated rejection. *Am J Transplant*. 2009;9(5):1099–1107.
121. Kakkar B, Makroo RN, Agrawal S, et al. Role of therapeutic plasma exchange in acute humoral rejection patients undergoing live-related renal transplantation: a single-center experience. *Asian J Transfus Sci*. 2021;15(1):62–67.
122. Gejyo F, Kawaguchi Y, Hara S, et al. Arresting dialysis-related amyloidosis: a prospective multicenter controlled trial of direct hemoperfusion with a beta2-microglobulin adsorption column. *Artif Organs*. 2004;28(4):371–380.
123. Tsuruya K, Arima H, Iseki K, Hirakata H, Kyushu Dialysis-Related Amyloidosis Study G. Association of dialysis-related amyloidosis with lower quality of life in patients undergoing hemodialysis for more than 10 years: the Kyushu Dialysis-Related Amyloidosis Study. *PLoS One*. 2021;16(8):e0256421.
124. Wagner B, Drel V, Gorin Y. Pathophysiology of gadolinium-associated systemic fibrosis. *Am J Physiol Renal Physiol*. 2016;311(1):F1–F11.
125. Poisson JL, Low A, Park YA. The treatment of nephrogenic systemic fibrosis with therapeutic plasma exchange. *J Clin Apher*. 2013;28(4):317–320.
126. Erkan D. Therapeutic and prognostic considerations in catastrophic antiphospholipid syndrome. *Autoimmun Rev*. 2006;6(2):98–103.
127. Bucciarelli S, Erkan D, Espinosa G, Cervera R. Catastrophic antiphospholipid syndrome: treatment, prognosis, and the risk of relapse. *Clin Rev Allergy Immunol*. 2009;36(2–3):80–84.
128. Clark WF, Rock GA, Buskard N, et al. Therapeutic plasma exchange: an update from the Canadian Apheresis Group. *Ann Intern Med*. 1999;131(6):453–462.
129. Gwathmey K, Balogun RA, Burns T. Neurologic indications for therapeutic plasma exchange: 2013 update. *J Clin Apher*. 2014;29(4):211–219.
130. Lehmann HC, Hartung HP, Hetzel GR, Stuve O, Kieseier BC. Plasma exchange in neuroimmunological disorders: part 2. Treatment of neuromuscular disorders. *Arch Neurol*. 2006;63(8):1066–1071.
131. Willison HJ, Jacobs BC, van Doorn PA. Guillain-Barre syndrome. *Lancet*. 2016;388(10045):717–727.
132. Randomised trial of plasma exchange, intravenous immunoglobulin, and combined treatments in Guillain-Barre syndrome. Plasma Exchange/Sandoglobulin Guillain-Barre Syndrome Trial Group. *Lancet*. 1997;349(9047):225–230.
133. van der Meche FG, Schmitz PI. A randomized trial comparing intravenous immune globulin and plasma exchange in Guillain-Barre syndrome. Dutch Guillain-Barre Study Group. *N Engl J Med*. 1992;326(17):1123–1129.
134. Syndrome TFCGoPEiG-B. Appropriate number of plasma exchanges in Guillain-Barre syndrome. *Ann Neurol*. 1997;41(3):298–306.
135. Koller H, Kieseier BC, Jander S, Hartung HP. Chronic inflammatory demyelinating polyneuropathy. *N Engl J Med*. 2005;352(13):1343–1356.
136. Brannagan 3rd TH. Current treatments of chronic immune-mediated demyelinating polyneuropathies. *Muscle Nerve*. 2009;39(5):563–578.
137. Conti-Fine BM, Milani M, Kaminski HJ. Myasthenia gravis: past, present, and future. *J Clin Invest*. 2006;116(11):2843–2854.

138. Garcia-Carrasco M, Escarcega RO, Fuentes-Alexandro S, Riebeling C, Cervera R. Therapeutic options in autoimmune myasthenia gravis. *Autoimmun Rev*. 2007;6(6):373–378.
139. Batocchi AP, Evoli A, Di Schino C, Tonali P. Therapeutic apheresis in myasthenia gravis. *Ther Apher*. 2000;4(4):275–279.
140. Pascuzzi RM. Myasthenia gravis and Lambert-Eaton syndrome. *Ther Apher*. 2002;6(1):57–68.
141. Stefanutti C, Julius U, Watts GF, et al. Toward an international consensus-Integrating lipoprotein apheresis and new lipid-lowering drugs. *J Clin Lipidol*. 2017;11(4):858–871.
142. Thompson GR, Group H-ULAW. Recommendations for the use of LDL apheresis. *Atherosclerosis*. 2008;198(2):247–255.
143. Kawashiri MA, Nohara A, Higashikata T, et al. Impact of evolocumab treatment on low-density lipoprotein cholesterol levels in heterozygous familial hypercholesterolemic patients withdrawing from regular apheresis. *Atherosclerosis*. 2017;265:225–230.
144. Jander S, Bischoff J, Woodcock BG. Plasmapheresis in the treatment of *Amanita phalloides* poisoning: II. A review and recommendations. *Ther Apher*. 2000;4(4):308–312.
145. Liu Y, Shu H, Long Y, et al. Development and internal validation of a Wasp Sting Severity Score to assess severity and indicate blood purification in persons with Asian wasp stings. *Clin Kidney J*. 2021;15(2):320–327.
- 145a. Maiwall R, Bajpai M, Singh A, et al. Standard-volume plasma exchange improves outcomes in patients with acute liver failure: A randomized controlled trial. *Clin Gastroenterol Hepatol*. 2022;20(4):e831–e854.
146. Hanish SI, Stein DM, Scalea JR, et al. Molecular adsorbent recirculating system effectively replaces hepatic function in severe acute liver failure. *Ann Surg*. 2017;266(4):677–684.
- 146a. Banares R, Nevens F, Larsen FS, et al. Extracorporeal albumin dialysis with the molecular adsorbent recirculating system in acute-on-chronic liver failure: the RELIEF trial. *Hepatology*. 2013;57(3):1153–1162.
147. Gerth HU, Pohlen M, Tholking G, et al. Molecular adsorbent recirculating system can reduce short-term mortality among patients with acute-on-chronic liver failure-A retrospective analysis. *Crit Care Med*. 2017;45(10):1616–1624.
148. Larsen FS, Schmidt LE, Bernsmeier C, et al. High-volume plasma exchange in patients with acute liver failure: an open randomised controlled trial. *J Hepatol*. 2016;64(1):69–78.
149. Hilal T, Morehead RS. Fulminant Wilson's disease managed with plasmapheresis as a bridge to liver transplant. *Case Rep Med*. 2014;2014:672985.
150. Ogura M, Makino H, Kamiya C, et al. Lipoprotein apheresis is essential for managing pregnancies in patients with homozygous familial hypercholesterolemia: seven case series and discussion. *Atherosclerosis*. 2016;254:179–183.
151. Bowman JM, Peddle LJ, Anderson C. Plasmapheresis in severe Rh iso-immunization. *Vox Sang*. 1968;15(4):272–277.
152. Scully M, Starke R, Lee R, Mackie I, Machin S, Cohen H. Successful management of pregnancy in women with a history of thrombotic thrombocytopenic purpura. *Blood Coagul Fibrinolysis*. 2006;17(6):459–463.
153. Guillevin L, Lhote F, Cohen P, et al. Corticosteroids plus pulse cyclophosphamide and plasma exchanges versus corticosteroids plus pulse cyclophosphamide alone in the treatment of polyarteritis nodosa and Churg-Strauss syndrome patients with factors predicting poor prognosis. A prospective, randomized trial in sixty-two patients. *Arthritis Rheum*. 1995;38(11):1638–1645.
154. Plumb LA, Oni L, Marks SD, Tullus K. Paediatric anti-neutrophil cytoplasmic antibody (ANCA)-associated vasculitis: an update on renal management. *Pediatr Nephrol*. 2018;33:25–39.
155. Hattori M, Ito K, Konomoto T, Kawaguchi H, Yoshioka T, Khono M. Plasmapheresis as the sole therapy for rapidly progressive Henoch-Schonlein purpura nephritis in children. *Am J Kidney Dis*. 1999;33(3):427–433.
156. Lee J, Clayton F, Shihab F, Goldfarb-Rumyantzev A. Successful treatment of recurrent Henoch-Schonlein purpura in a renal allograft with plasmapheresis. *Am J Transplant*. 2008;8(1):228–231.